



Rigorous Text Analysis in Innovation Research

Practices, priorities, and rigor strategies

Vito Giordano

Assistant Professor at University of Pisa

AGENDA

A circular inset image showing a lecture hall. A presenter stands at the front near a screen displaying 'I Grafi della Cor' and 'Graph Learning Suggestions in P'. An audience is seated in the foreground, viewed from behind.

01 INTRODUCTION

02 TEXT AS RESEARCH METHOD

03 VALIDITY AND RELIABILITY

04 RIGOR IN MANAGEMENT



01 INTRODUCTION

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04 RIGOR IN MANAGEMENT

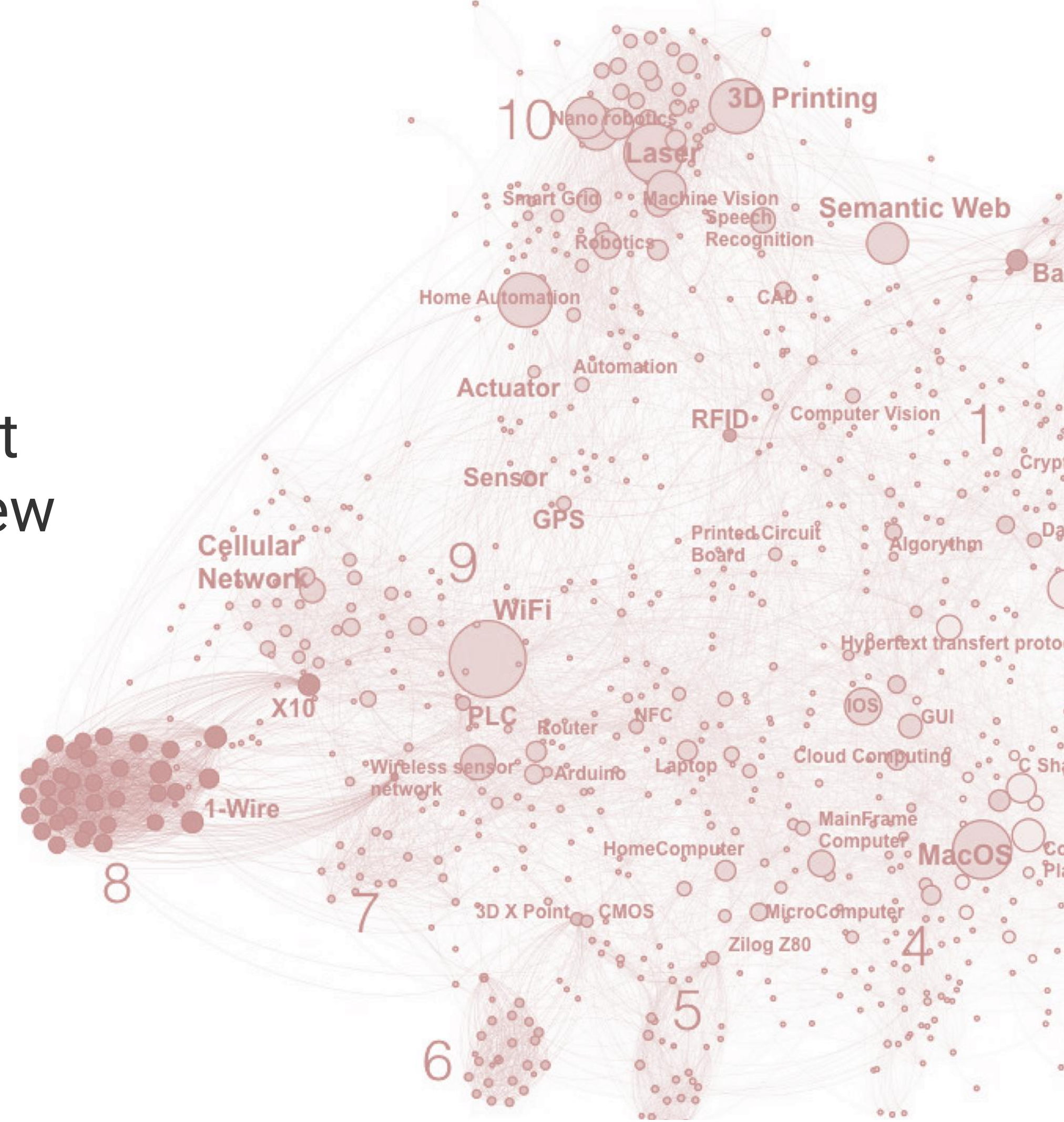
INTRODUCTION

Text as Data

Management is highly **knowledge-intensive processes**. They generate a vast amount of expert knowledge, and it requires expert knowledge to produce new insights.

Around 80% of data in organizations is estimated to consist of text.

Hickman, L., Thapa, S., Tay, L., Cao, M., & Srinivasan, P. (2022). Text preprocessing for text mining in organizational research: Review and recommendations. Organizational Research Methods, 25(1), 114-146.

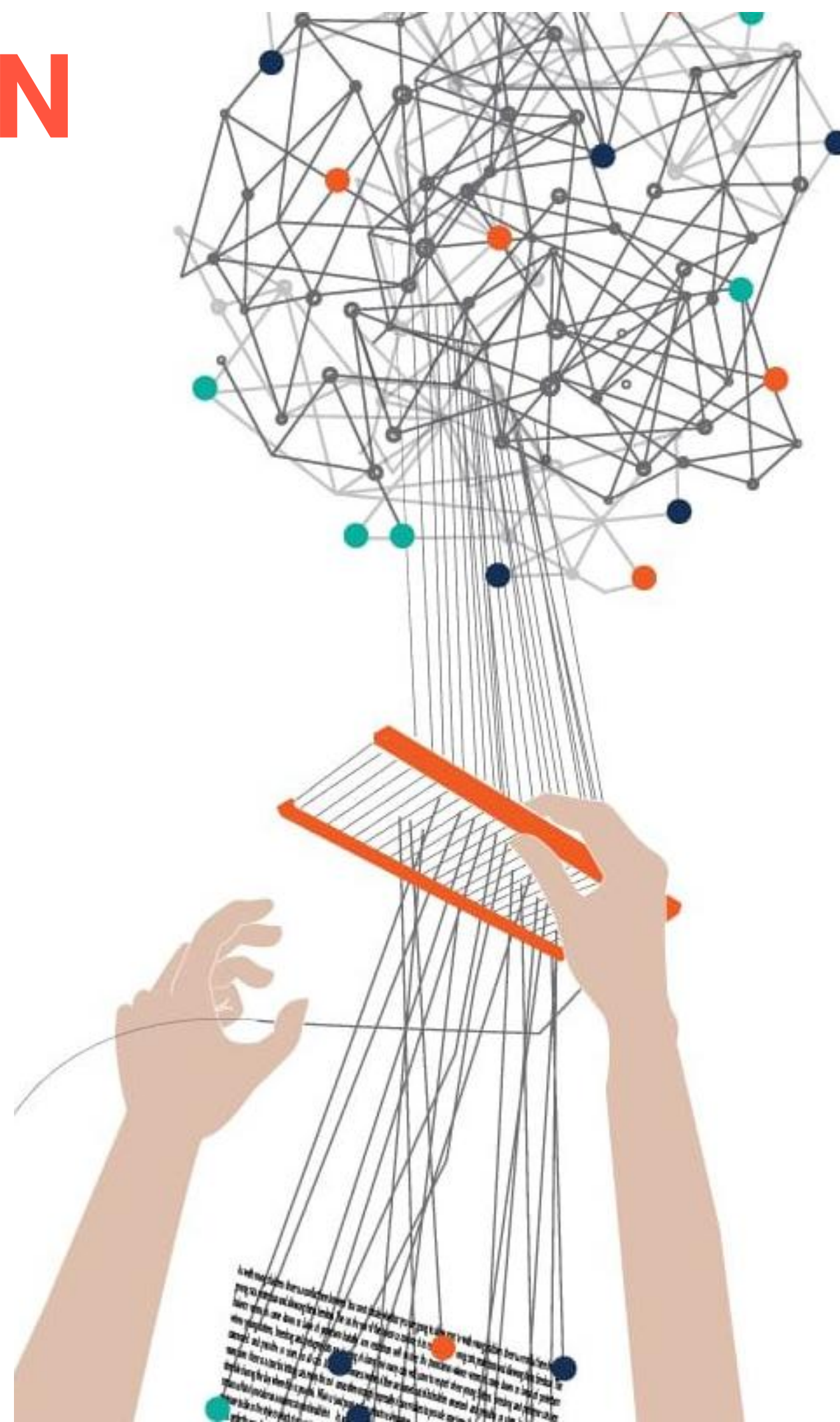


INTRODUCTION

Text Analysis

The use of **text** as **data** opens **new opportunities** for Management researchers to study phenomena that were previously difficult to investigate using structured data alone in **more reliable way**.

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Text develops theory

01

Makes previously elusive constructs **measurable**.

Text validates theory

02

A language-based measure must capture the construct it claims to capture. And must **predict** what the theory says it should.

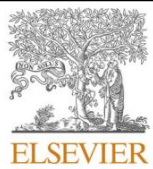
INTRODUCTION

Why in Innovation Management

J PROD INNOV MANAG 2013;30(6):1099–1111
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DOI: 10.1111/jpim.12048

A Review of Longitudinal Research in the Product Innovation Field, with Discussion of Utility and Conduct of Sequence Analysis

Helen Perks and Deborah Roberts



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Case study research on innovation systems: Paradox, dialectical analysis and resolution

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Ethnographies in the Front End: Designing for Enhanced Customer Experiences*

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& Management

Perspective: State-of-the-Art: The Quality of Case Study Research in Innovation Management*

Keith Goffin, Pär Åhlström, Mattia Bianchi and Anders Richtnér

CATALYST

Same model, same data, but different outcomes: Evaluating the impact of method choices in structural equation modeling

Marko Sarstedt^{1,2} | Susanne J. Adler¹ | Christian M. Ringle^{3,4} | Gyeongcheol Cho⁵ | Adamantios Diamantopoulos⁶ | Heungsun Hwang⁷ | Benjamin D. Lienggaard⁸

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PRODUCT
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MANAGEMENT

Twenty Years of the *Journal of Product Innovation Management*: History, Participants, and Knowledge Stock and Flows

Wim Biemans, Abbie Griffin, and Rudy Moenaert

Are rigor and transparency enough? Review and future directions for case studies in technology and innovation Management

Ziad Elsahn¹, Lisa Callagher², Kenneth Husted², Stefan Korber² and Frank Siedlok²

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There is no shared understanding within the innovation research community of what constitutes a “high-quality” text analysis output. So what is a **valid finding** and how to **validate the findings** obtained with text analysis is underexplored.

Antons, D., Grünwald, E., Cichy, P., & Salge, T. O. (2020). *The application of text mining methods in innovation research: current state, evolution patterns, and development priorities. R&D Management*, 50(3), 329-351.

Humphreys, A., & Wang, R. J. H. (2018). *Automated text analysis for consumer research. Journal of Consumer Research*, 44(6), 1274-1306.

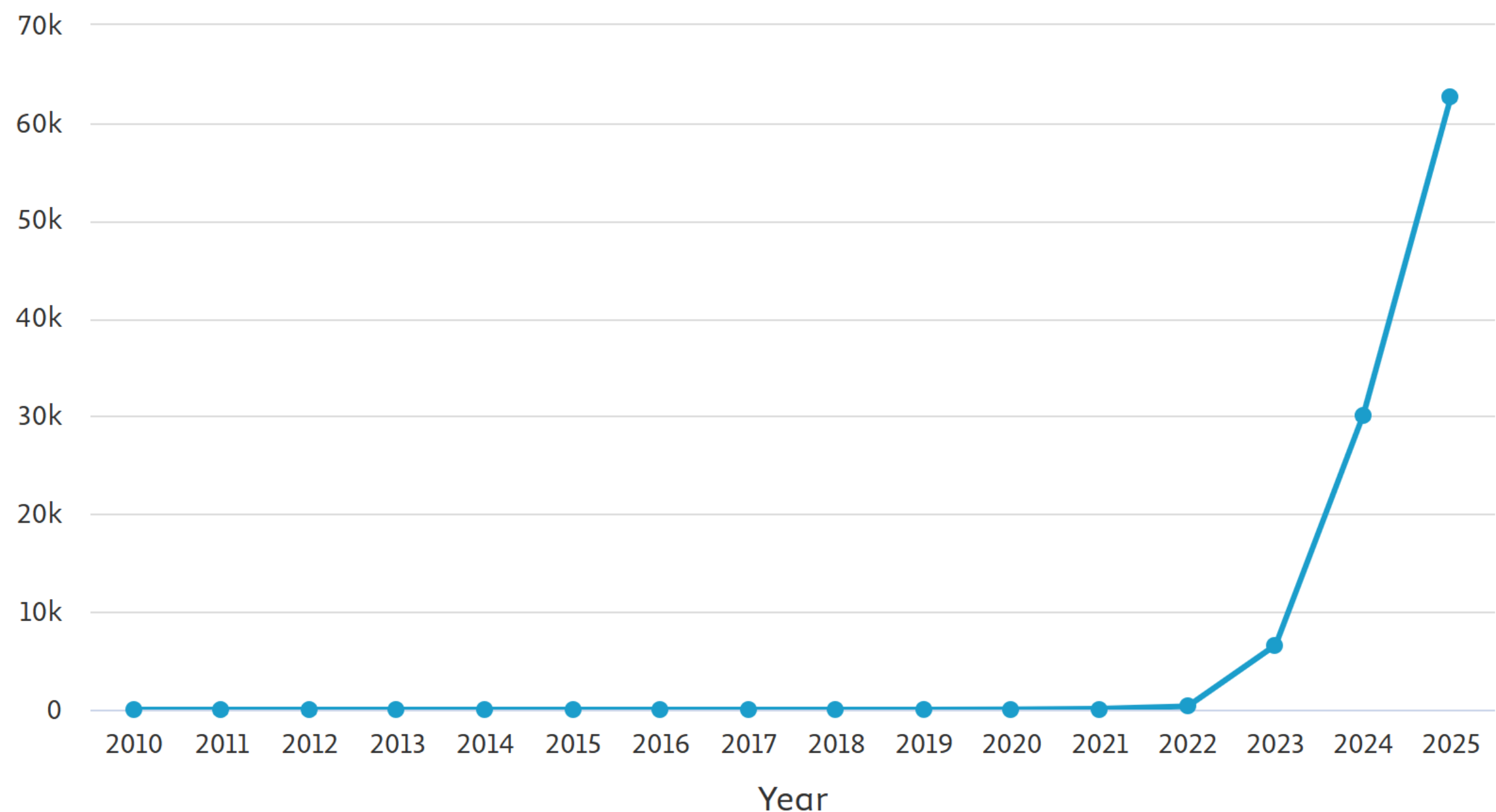
Xu, H., Zhang, N., & Zhou, L. (2020). *Validity concerns in research using organic data. Journal of Management*, 46(7), 1257-1274.

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LLMs significantly **lower the technical barriers** in the extraction of information from text, leading to **loss of methodological control**

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INTRODUCTION

Why in Innovation Management

	Rigor Dimension	Method	Field
<i>Xu, H., Zhang, N., & Zhou, L. (2020). Validity concerns in research using organic data. Journal of Management, 46(7), 1257-1274.</i>	Validity	Computational	Management
<i>Short, J. C., Broberg, J. C., Coglisier, C. C., & Brigham, K. H. (2010). Construct validation using computer-aided text analysis (CATA) an illustration using entrepreneurial orientation. Organizational research methods, 13(2), 320-347.</i>	Construct Validity	Dictionary-based	Management
<i>Belderbos, R., Grabowska, M., Leten, B., Kelchtermans, S., & Ugur, N. (2017). On the use of computer-aided text analysis in international business research. Global Strategy Journal, 7(3), 312-331.</i>	Construct Validity	Dictionary-based	International business
<i>Liu, S., Hickman, L., Dahlander, L., & Piezunka, H. (2026). Textual Similarity in Organizational Research: Review of Applications, Consistency of Methods, and Best Practice Recommendations. Organizational Research Methods, 10944281261432629.</i>	Validity	Text Similarity	Management
<i>Berger, J., Humphreys, A., Ludwig, S., Moe, W. W., Netzer, O., & Schweidel, D. A. (2020). Uniting the tribes: Using text for marketing insight. Journal of marketing, 84(1), 1-25.</i>	Validity & Reliability	Text Analysis	Marketing
<i>Campion, E. D., & Campion, M. A. (2025). A review of text analysis in human resource management research: Methodological diversity, constructs identified, and validation best practices. Human Resource Management Review, 35(2), 101078.</i>	Validity & Reliability	Text Analysis	Human Resources Management

INTRODUCTION

Why in Innovation Management



Wetzels, M., Wetzels, R., Schweiger, E., & Grewal, D. (2025). How Topic Modeling Can Spur Innovation Management. *Journal of Product Innovation Management*.

ORIGINAL ARTICLE OPEN ACCESS

How Topic Modeling Can Spur Innovation Management

Martin Wetzels¹ | Ruud Wetzels² | Elisa Schweiger³ | Dhruv Grewal^{4,5,6}

¹EDHEC Business School, Lille Campus, Roubaix, France | ²The University of Sydney Business School, The University of Sydney, Sydney, Australia | ³King's College, London, UK | ⁴Babson College, Boston, USA | ⁵University of Bath, Bath, UK | ⁶Tecnológico de Monterrey, Monterrey, Mexico

Correspondence: Ruud Wetzels (ruud.wetzels@sydney.edu.au)

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Keywords: innovation processes | topic modeling | topical evolution | topical impact | unstructured text data

ABSTRACT

The avalanche of available unstructured text data makes it ever more challenging for innovation practitioners (and academics) to extract meaningful insights from such data. Topic modeling can support these efforts and help spur innovation. The current study reviewed 1099 innovation management articles to identify and compare the most frequently used probabilistic topic modeling approaches for innovation. In an effort to contextualize the suitability of these approaches, we develop a framework to organize existing topic modeling applications along the different innovation stages (i.e., idea generation, development, and commercialization) and innovation research in general. By zooming in on the three innovation stages, the authors showcase how topic modeling can spur innovation within each stage and highlight the future potential of the specific approaches. To further assist in capturing the various dynamics in complex unstructured text datasets, we illustratively apply a tailored topic modeling configuration to 1444 *Journal of Product Innovation Management* articles (1984–2023) to identify emerging, stable, and mature topics, as well as looking at their respective impact. This demonstration could serve as a starting point or blueprint for innovation practitioners and researchers seeking to combine the advantages of several topic modeling approaches. We conclude by offering a future outlook, including a forward-looking research agenda. Taken together, our study offers guidance to and equips innovation practitioners and academics to design distinctive topic modeling procedures to best serve their intended purposes. If deployed appropriately, topic modeling helps users extract a wealth of unique, unprecedented insights from a continuously expanding source of data.

1 | Introduction

In the innovation domain, the incredible paradigm shift toward digital innovation management has involved the introduction of novel digital business models, ever-increasing proliferation of

tools accordingly have been introduced to take advantage of the unique properties and increasing availability of (mainly) digital, unstructured textual data (Cooper 2021; Oracle 2024).

Natural language processing (NLP), which relies on computa-

VALIDITY AND RELIABILITY

Text Analysis and the problem of Rigor

We have the result of a brainstorming session



“Smartphone case that physically blocks access to social media apps during study or work time.”

“Put the phone inside a box, set the time, and the box does not open until the work session is finished.”
The cover has a small timer on it. When the timer starts, social media apps cannot be opened.

“When the user wears the bracelet, social media apps are automatically blocked on the phone.”
A group of friends starts a challenge. If someone opens social media, everyone can see it, and that person loses points.

“When the lamp is turned on for studying, it sends a signal to the phone to block Instagram, TikTok, Facebook, and other apps.”
Students tap a card on the phone before class, and the phone blocks social media until the lesson ends.

“When the lamp is turned on for studying, it sends a signal to the phone and blocks Instagram, TikTok, Facebook, and similar apps.”
A reward system for not using social media

“One big button activates a mode where only calls, calendar, and useful apps work. Social media disappears for some hours.”
A device at home where people place their phones during dinner, study, or family time. The device blocks notifications and social media.

“A group of friends starts a challenge. If someone opens social media, everyone can see it, and that person loses points.”

VALIDITY AND RELIABILITY

Text Analysis and the problem of Rigor

How can we **measure** whether the idea is **novel**?



💡 *HINT: Word Embeddings*

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VALIDITY AND RELIABILITY

Text Analysis and the problem of Rigor

Scale used: 0 = essentially already exists; 10 = highly original. I scored **novelty**, not market value or feasibility. Existing prior art includes Forest’s digital-tree mechanic, timed phone lock boxes, NFC/card-based blockers such as Brick/Bloom/Foqos, and standard app-blocker/focus-mode tools. 🌳 Forest +4



#	Idea	Novelty	Why
1	Friends can see if someone opens social media; person loses points	5/10	Social accountability + penalty is moderately novel, though gamified focus and accountability systems are common.
2	Study lamp turns on and blocks Instagram/TikTok/Facebook	7/10	The app-blocking part is common, but using a lamp as the physical trigger is more distinctive.
3	Reward system for not using social media	2/10	Very common gamification pattern.
4	Students tap a card before class; phone blocks social media until lesson ends	3/10	Similar to NFC/card app blockers already on the market. 📱 App Store +1
5	Put phone in timed box that will not open	1/10	Timed phone lock boxes already exist. 🛒 Amazon +1
6	Wearing a bracelet automatically blocks social media apps	7/10	Wearable-triggered blocking is less common than NFC cards or apps, so the trigger mechanism feels more novel.
7	Study lamp blocks social media apps	7/10	Duplicate of #2.

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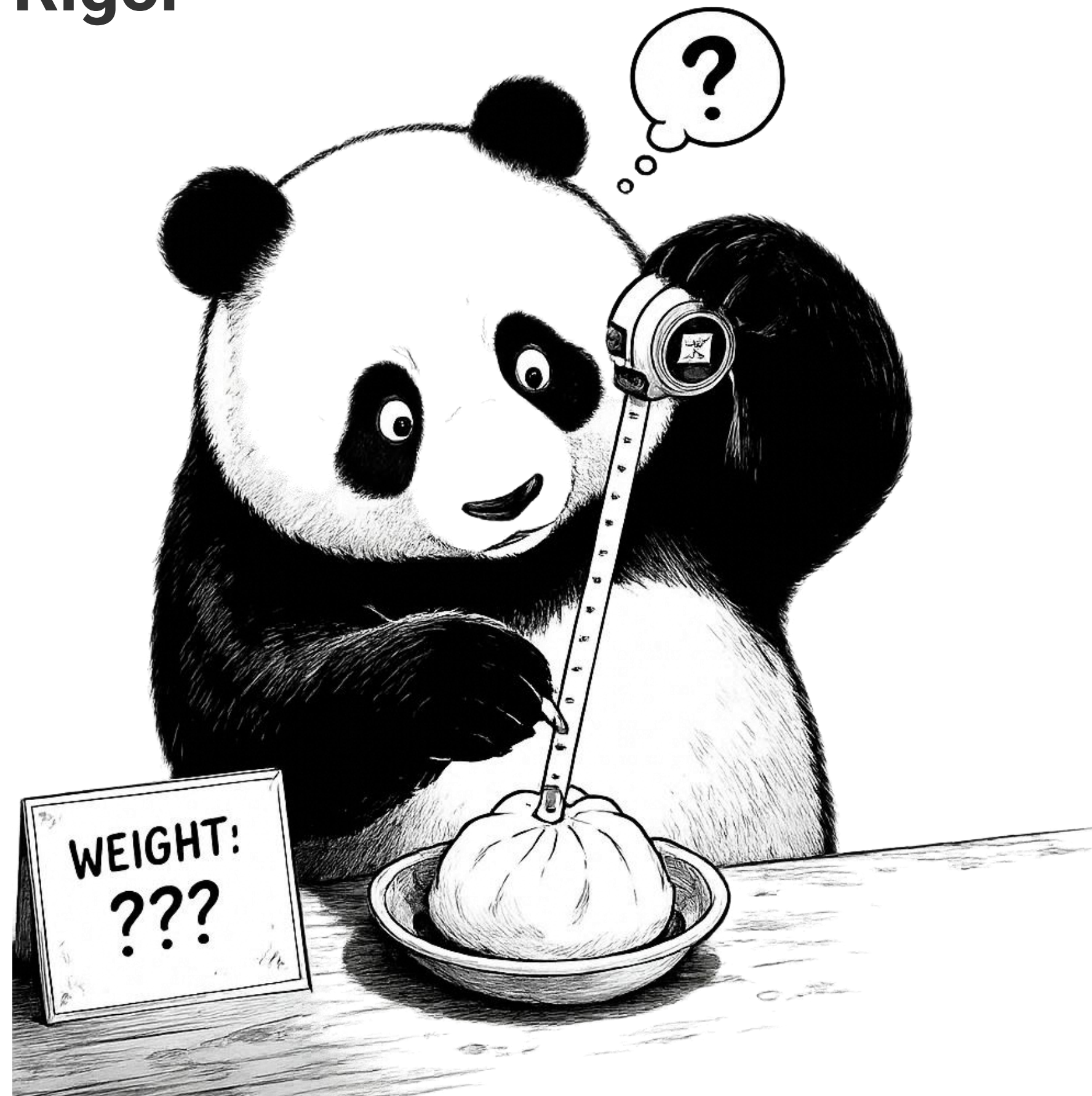
- How can we know whether the **output** generated by a LLM is **valid**?
- And how can we know whether **repeating the same prompt** will produce the **same output**?



#	Idea	Novelty	Why
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INTRODUCTION

Rigor



Several scholars emphasised concerns about **methodological rigor in relation to text analysis**.

Methodological concerns, if not addressed, **may threaten the validity and reliability** of the research.

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Günther, W. A., Joshi, M., & Constantinides, P. (2026). Trustworthiness in Computational Theory Construction: Dimensionalization and Category Surfacing. MIS Q, 1-30.

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What are the implications when the text analysis used to generate evidence **lacks validity and reliability?**



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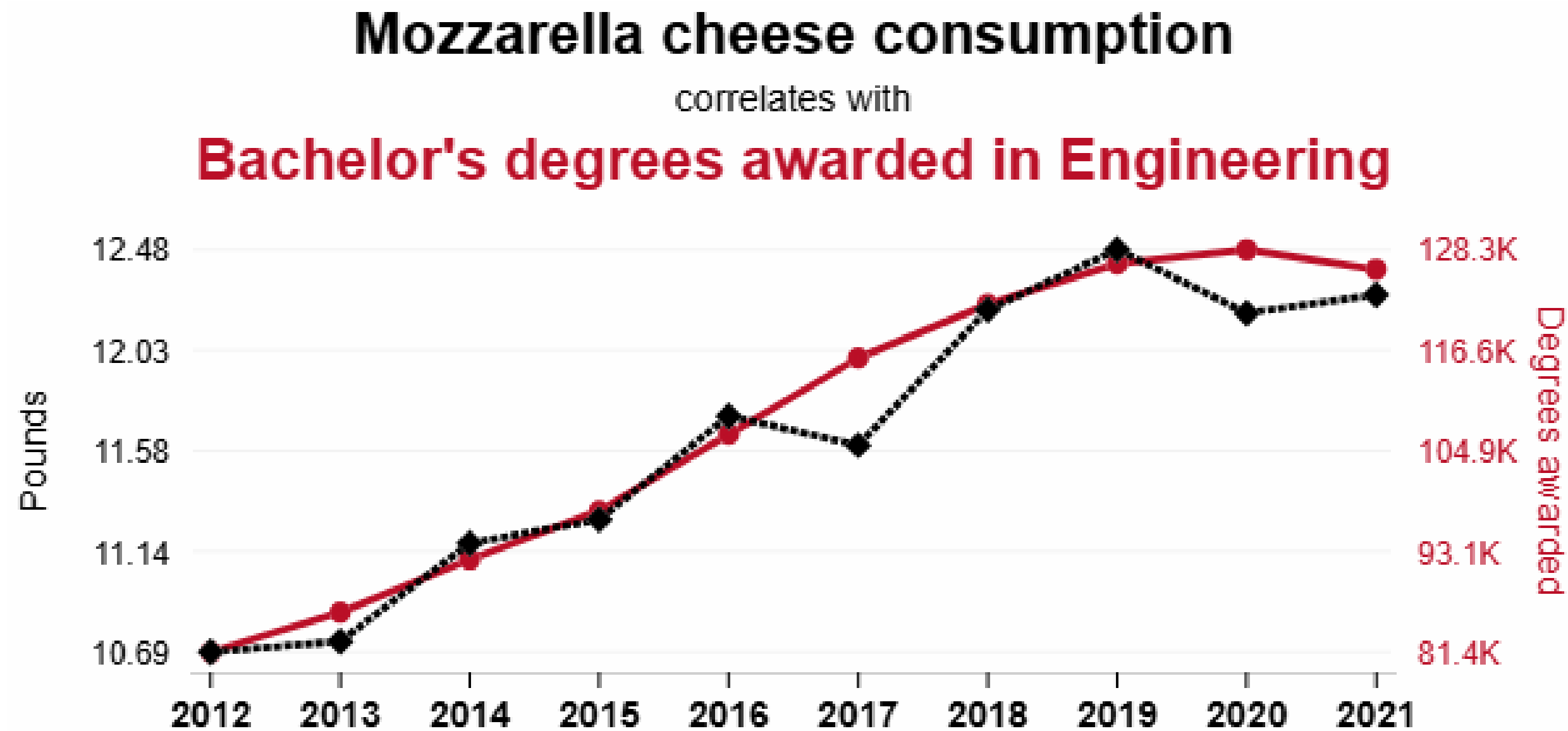
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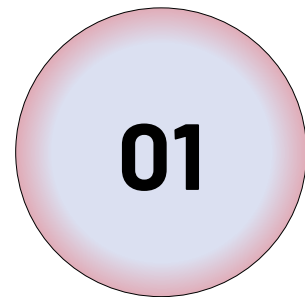
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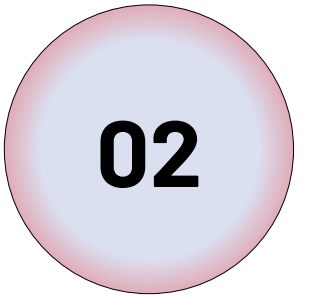


Theoretical Impact

Reduces the contribution of the study to cumulative management knowledge

Practical Impact

May lead to misleading managerial or policy implications



INTRODUCTION

Rigor

01

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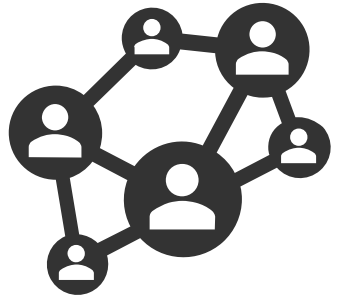
02

VALIDITY AND RELIABILITY

How can we address this problem?

Establish a **common dictionary** of key concepts related to rigor

1



Understand the **specific characteristics of text analysis** as research method

2



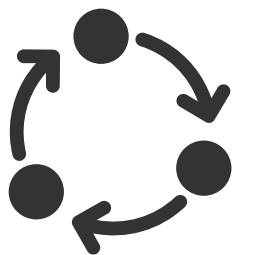
Examine how rigor strategies are applied in management using text analysis

3



Develop ad hoc methods to ensure rigor in text analysis

4





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02 TEXT AS RESEARCH METHOD

03 VALIDITY AND RELIABILITY

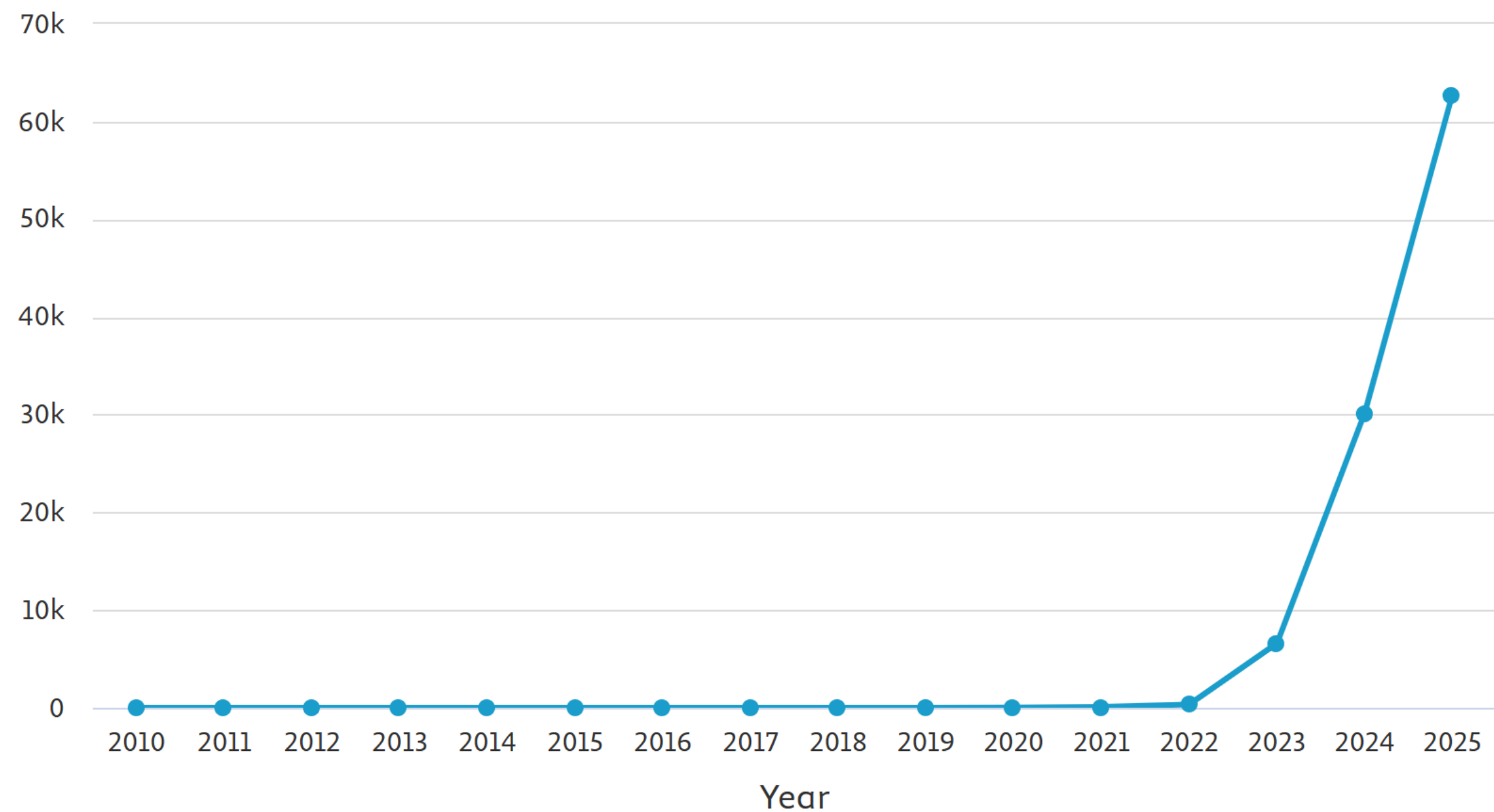
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TEXT AS RESEARCH METHOD

Text Analysis as a Research Method

TITLE-ABS-KEY ("generative artificial intelligence" OR "generative ai" OR "large language model")

99,983 document results

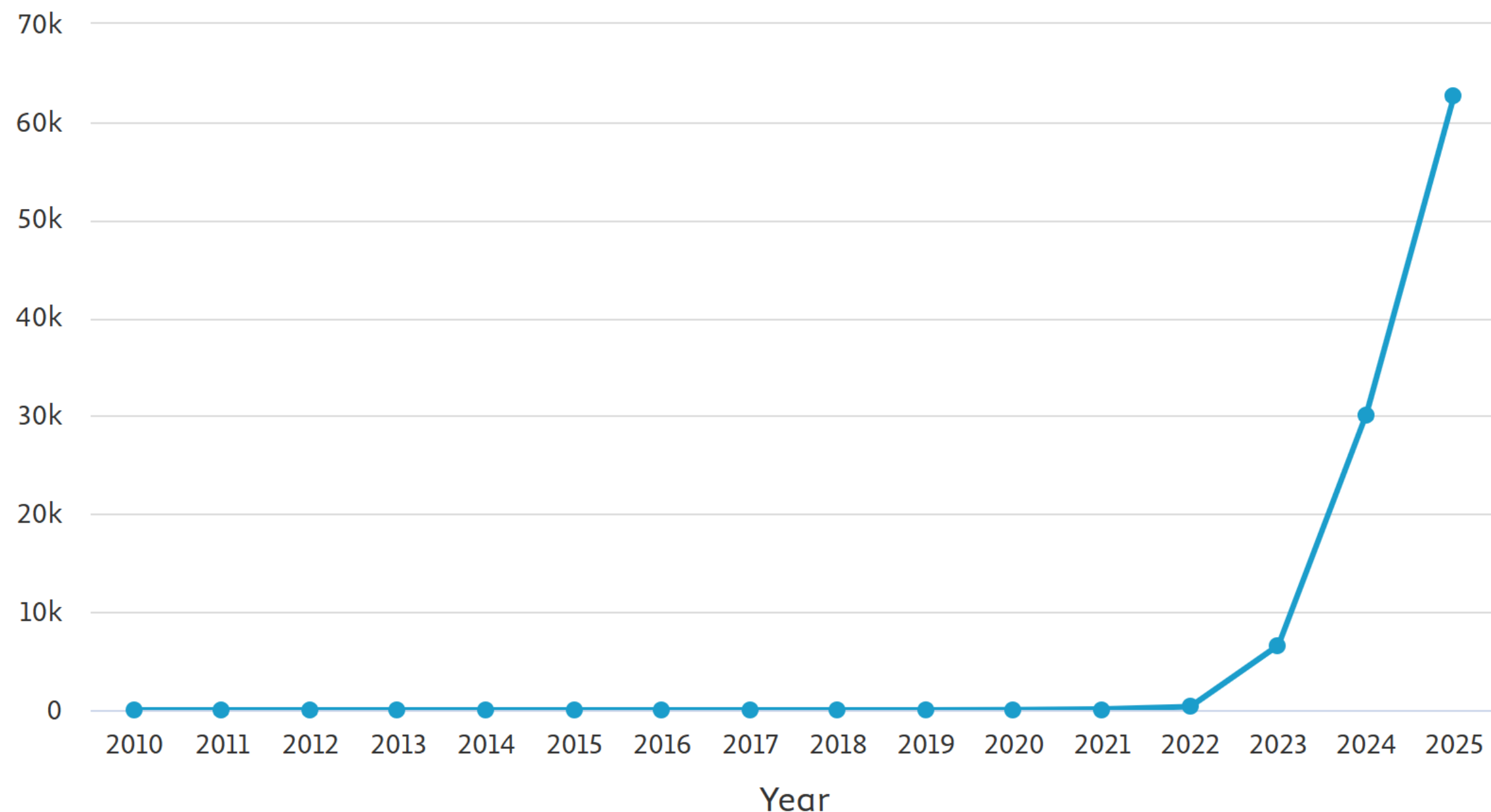


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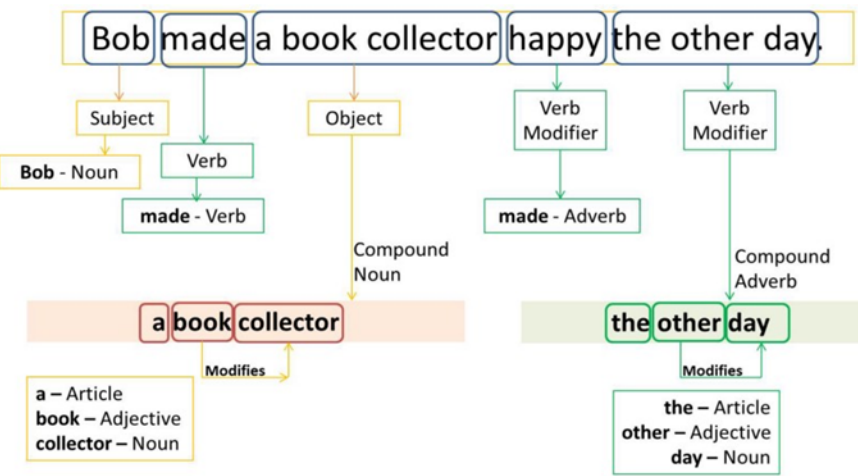
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TEXT AS RESEARCH METHOD

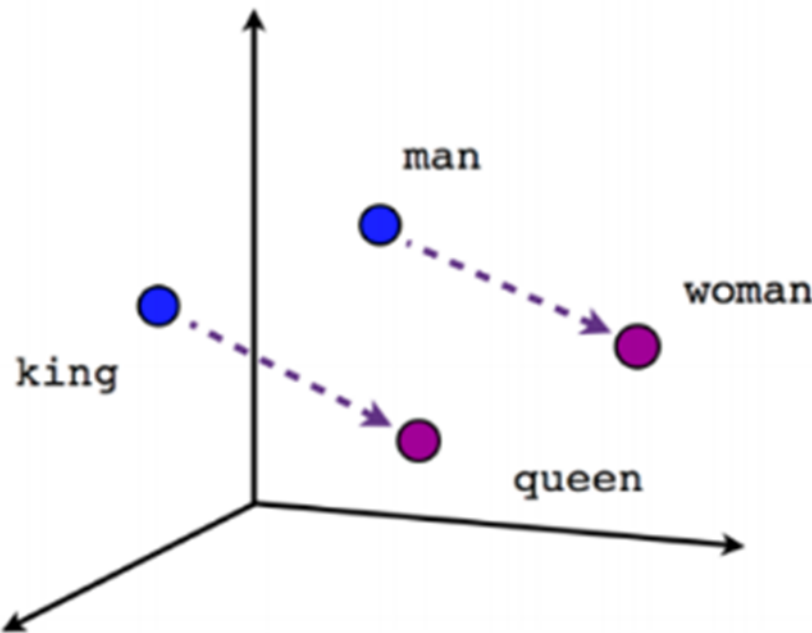
Text Analysis Pipeline



This phase focuses on cleaning and standardizing the text to **reduce noise**. By removing irrelevant elements and harmonizing formats, preprocessing improves data quality and consistency. In addition, it helps manage the high dimensionality of textual data.



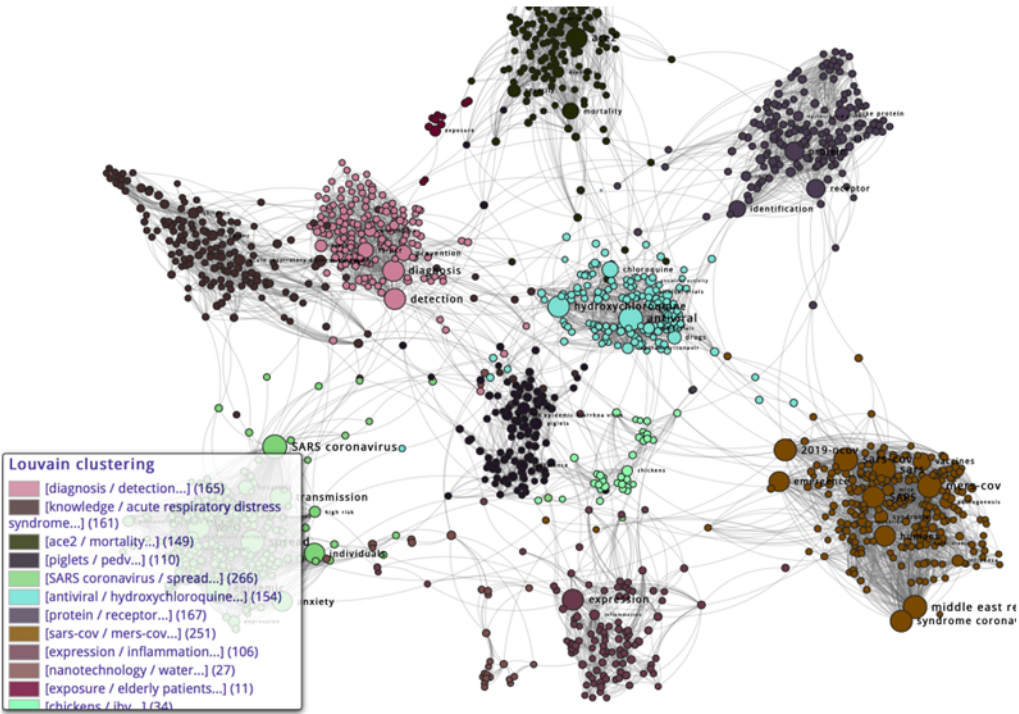
Text representation refers to the process of converting raw textual data into structured, **machine-readable formats** that can be analyzed computationally.



Text modelling uses computational methods to extract information from text. It can be **supervised**, when models learn from labeled data to identify predefined categories or constructs, or **unsupervised**, when models explore the text to uncover patterns such as topics or clusters without prior labels.

tourists, the third ORDINAL most visited in the European Union ORG, and the most popular tourist destination. [13] Its historic centre is listed by UNESCO ORG as a World Heritage ORG Site.[14] Host city for the Summer Olympics EVENT, Rome GPE is also the seat of several specialised agencies of the United ORG, such as the Food and Agriculture Organization ORG (FAO ORG), the World Food Programme ORG and the International Fund for Agricultural Development ORG (IFAD ORG). The city also hosts the Secretariat of the Parliamentary Assembly ORG of the Union for the Mediterranean[15] (UfM ORG) as headquarters of many international business companies such as Eni ORG, Enel GPE, TIM ORG, S.p.A. ORG, and national and international banks such as Unicredit ORG and BNL ORG. Rome GPE business district is the home of many companies involved in the oil industry, the pharmaceutical industry services. The presence of renowned international brands in the city have made Rome GPE an important centre.

Outputs of Text modelling are subsequently used as inputs for **downstream analyses**.



Feuerriegel, S., Maarouf, A., Bär, D., Geissler, D., Schweisthal, J., Pröllochs, N., ... & Van Bavel, J. J. (2025). Using natural language processing to analyse text data in behavioural science. *Nature Reviews Psychology*, 4(2), 96-111.

TEXT AS RESEARCH METHOD

Text Analysis

Supervised Modelling

Dictionary-based

Representation-based machine learning

Large Language Models

Unsupervised Modelling

Topic Modelling

Text Clustering

Text Similarity

Kobayashi, V. B., Mol, S. T., Berkers, H. A., Kismihók, G., & Den Hartog, D. N. (2018). Text mining in organizational research. Organizational research methods, 21(3), 733-765.

Berger, J., Humphreys, A., Ludwig, S., Moe, W. W., Netzer, O., & Schweidel, D. A. (2020). Uniting the tribes: Using text for marketing insight. Journal of marketing, 84(1), 1-25.

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TEXT AS RESEARCH METHOD

Supervised Modelling

IN THE MATTER OF a proposed contract between the Department of Citywide Administrative Services **ORG** of the City of New York **GPE** and Tesla, Inc. **ORG** , located at 3500 **CARDINAL** Deer Creek Rd., Palo Alto **GPE** , CA 94304, for procuring Tesla Model 3 All-Electric Sedans. The contract is in the amount of \$ 12,360,000.00 **MONEY** . The term of the contract shall be five years from date **DATE** of Notice of Award. The proposed contractor has been selected by Sole Source Procurement Method **ORG** , pursuant to Section 3-05 **LAW** of the Procurement Policy Board Rules **ORG** . If the plan does go through, the \$12.36 million **MONEY** could effectively purchase about 274 **CARDINAL** units of the base Model 3 Rear-Wheel-Drive, which cost \$ 44,990 **MONEY** under Tesla **ORG** 's current pricing structure.

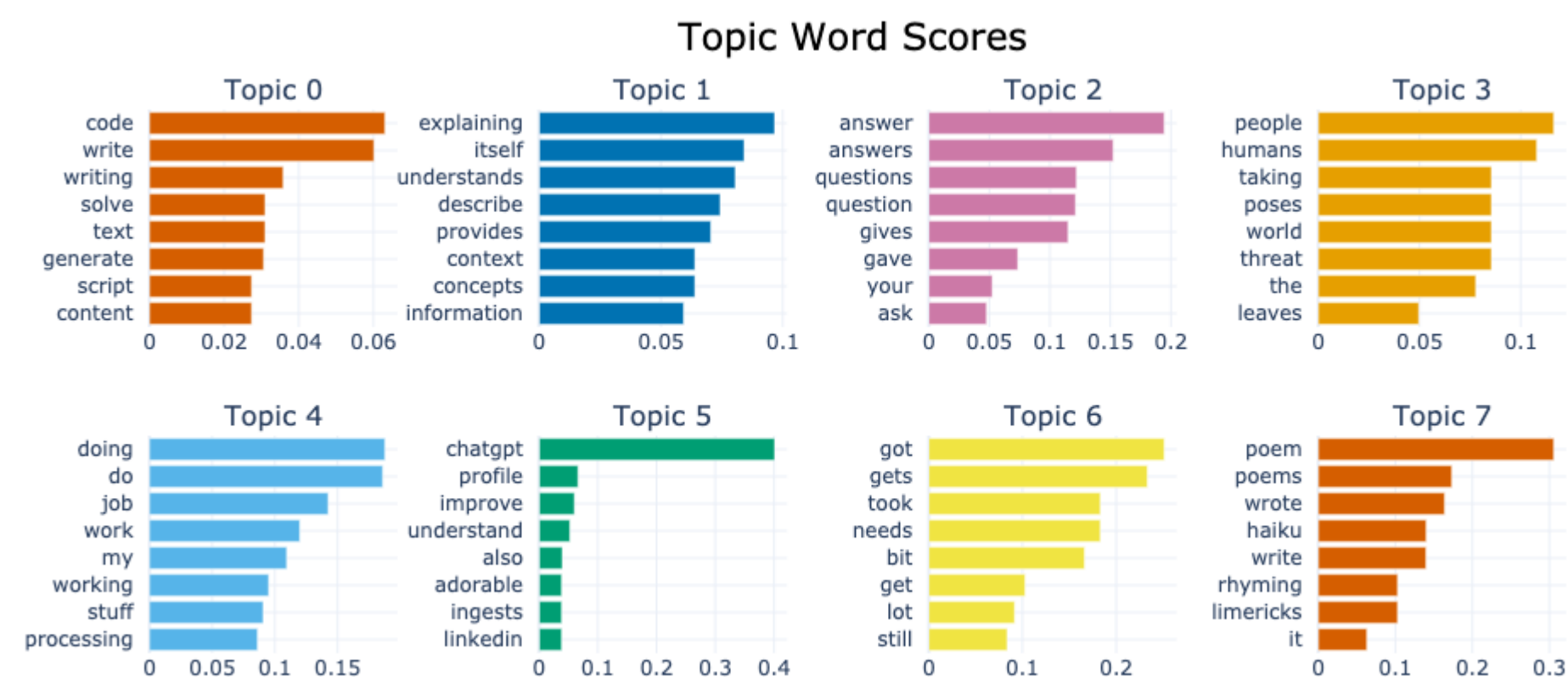
Word-level classification

Loves the German bakeries in Sydney. Together with my imported honey it feels like home	Positive
@VivaLaLauren Mine is broken too! I miss my sidekick	Negative
Finished fixing my twitter...I had to unfollow and follow everyone again	Negative
@DinahLady I too, liked the movie! I want to buy the DVD when it comes out	Positive
@frugaldougal So sad to hear about @OscarTheCat	Negative
@Mofette brilliant! May the fourth be with you #starwarsday #starwars	Positive
Good morning thespians a bright and sunny day in UK, Spring at last	Positive
@DowneyisDOWNEY Me neither! My laptop's new, has dvd burning/ripping software but I just can't copy the files somehow!	Negative

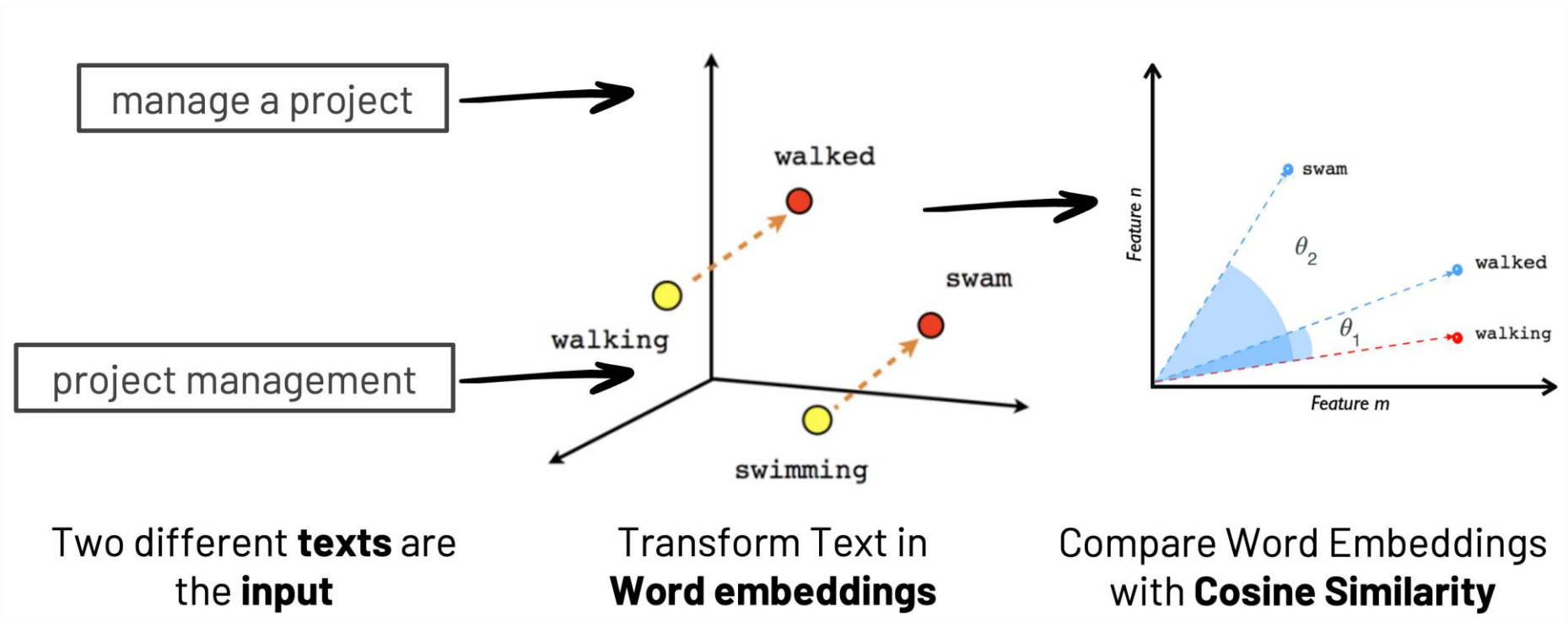
Sentence-level classification

TEXT AS RESEARCH METHOD

Unsupervised Modelling



Topic Modelling

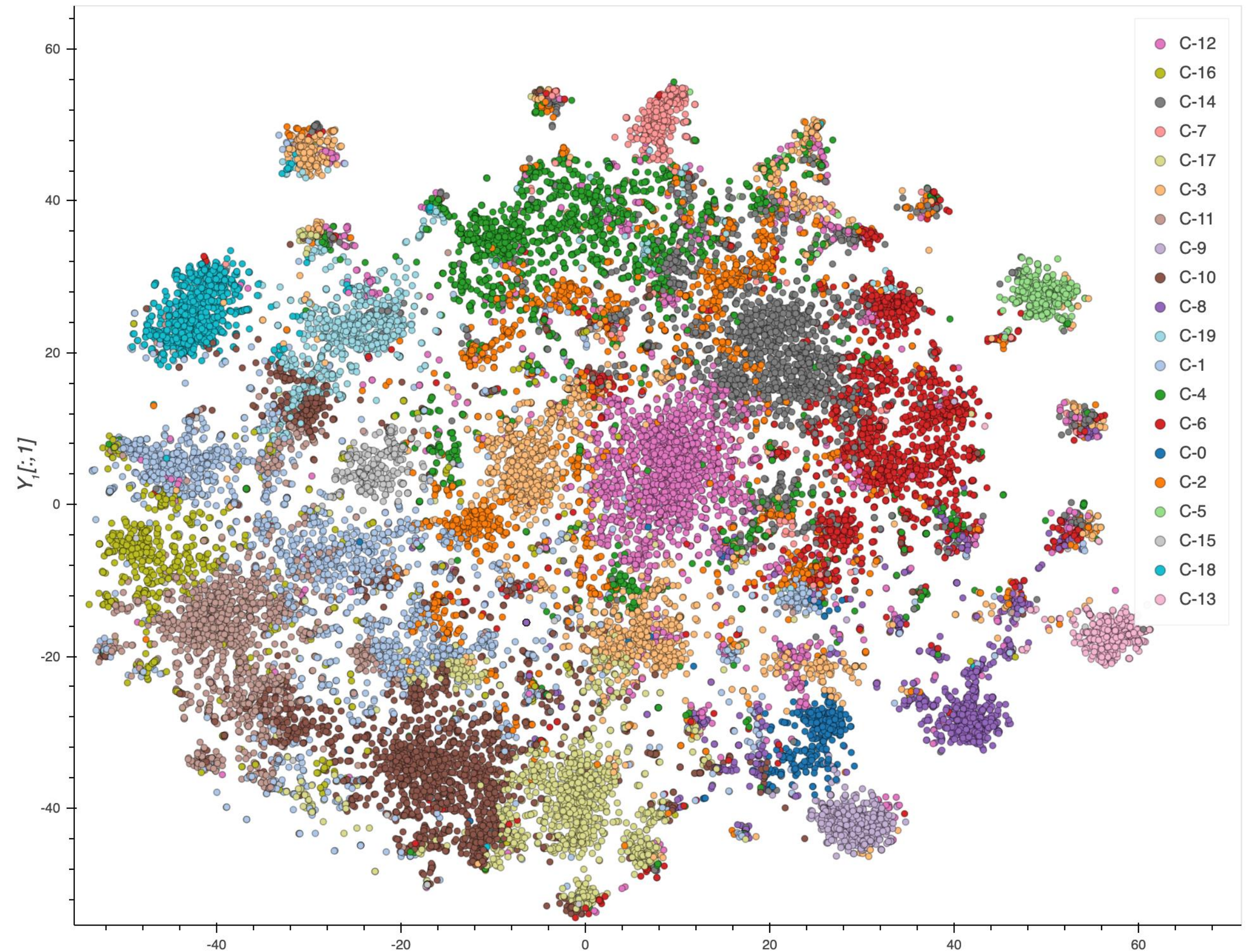


Text Similarity

TEXT AS RESEARCH METHOD

Unsupervised Modelling

Text Clustering





01 INTRODUCTION

02 TEXT AS RESEARCH METHOD

03 VALIDITY AND RELIABILITY

04 RIGOR IN MANAGEMENT

VALIDITY AND RELIABILITY

Common dictionary

Establish a **common dictionary** of key concepts related to rigor

1

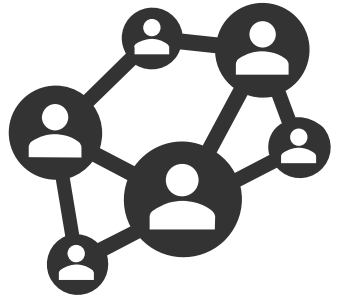


VALIDITY AND RELIABILITY

Common dictionary

Establish a **common dictionary** of key concepts related to rigor

1



Validity

Validity refers to the approximate **truth** of an inference

Shadish, W.R., Cook, T.D. and Campbell, D.T. (2002) Experimental and Quasi-Experimental Designs for Generalized Causal Inference. Houghton, Mifflin and Company.

Reliability

Reliability is the extent to which the **same observational procedure in the same context yields the same information.**

Kirk, J., & Miller, M. L. (1986). Reliability and validity in qualitative research (Vol. 1). Sage.

VALIDITY AND RELIABILITY

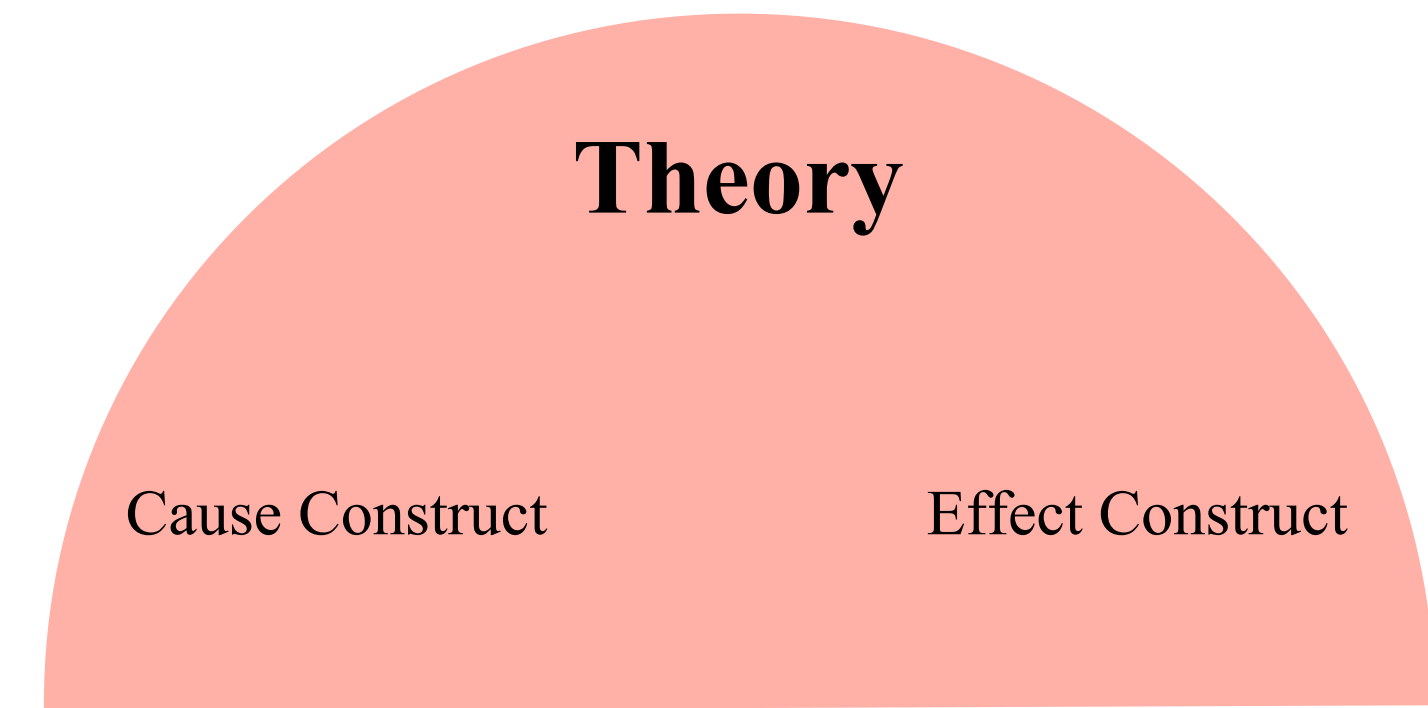
Common dictionary

Text Analysis can be used to **operationalize** and **measure** management **constructs** by extracting structured indicators from unstructured textual data.

VALIDITY AND RELIABILITY

Common dictionary

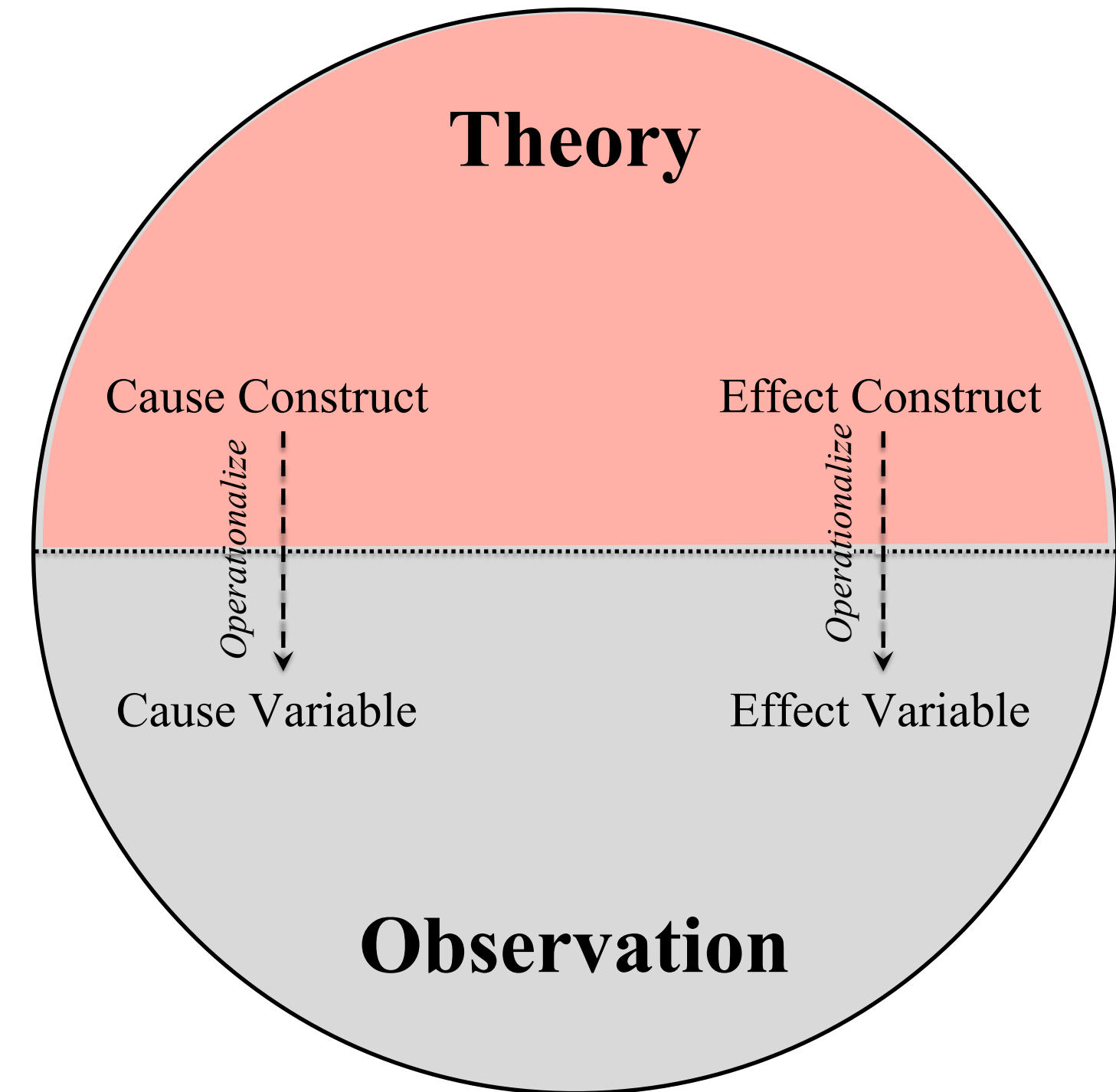
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VALIDITY AND RELIABILITY

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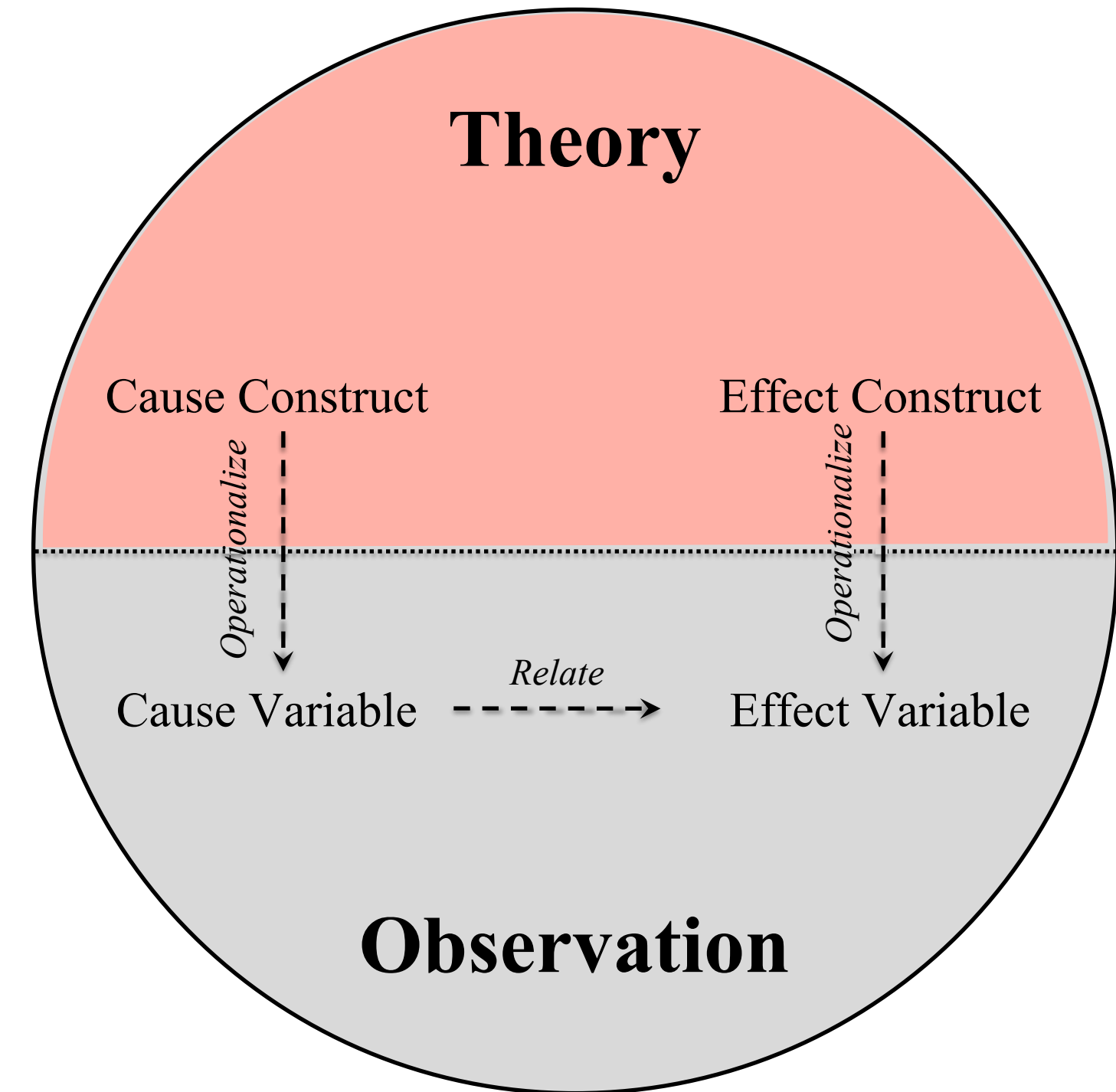
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VALIDITY AND RELIABILITY

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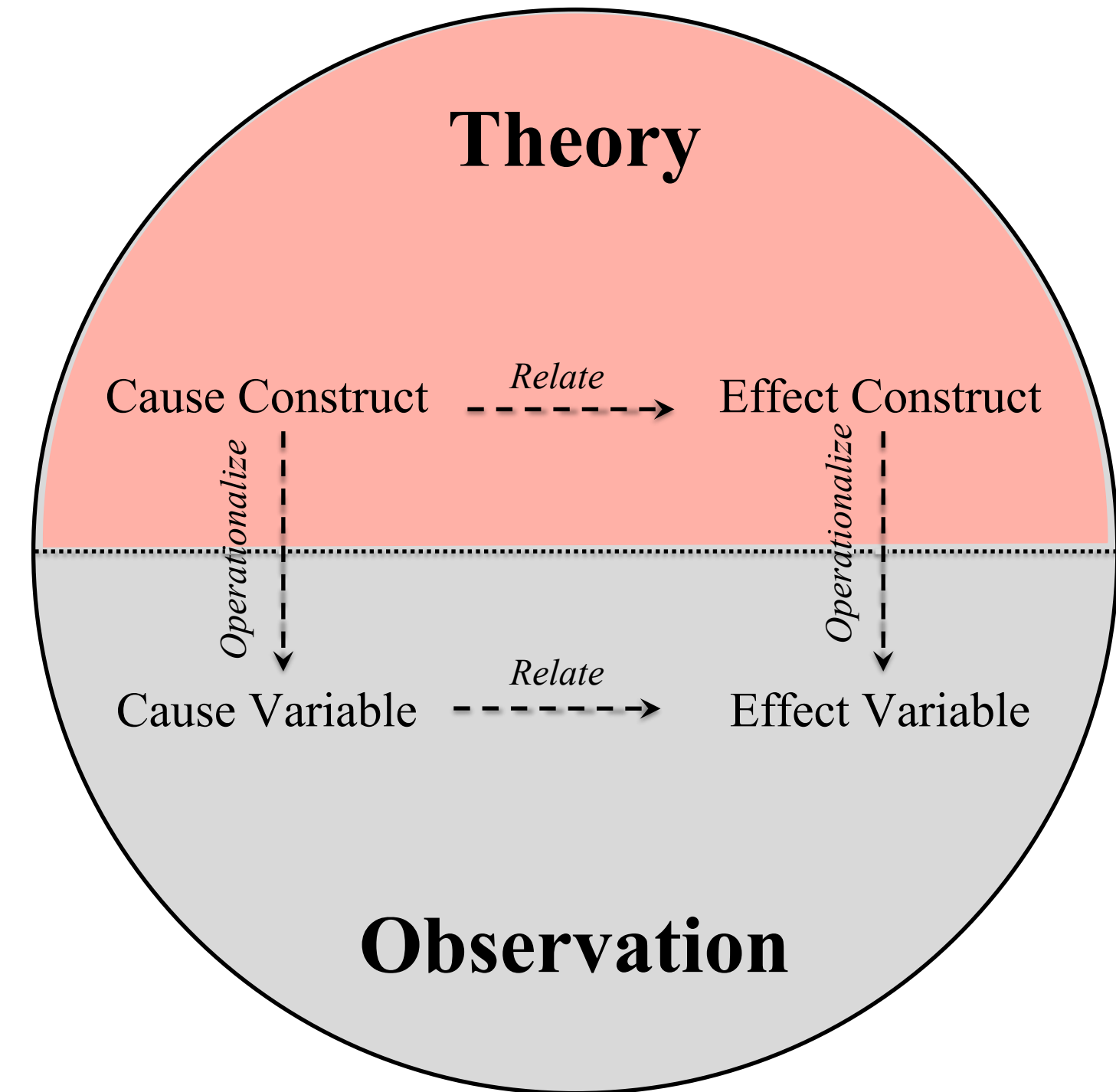
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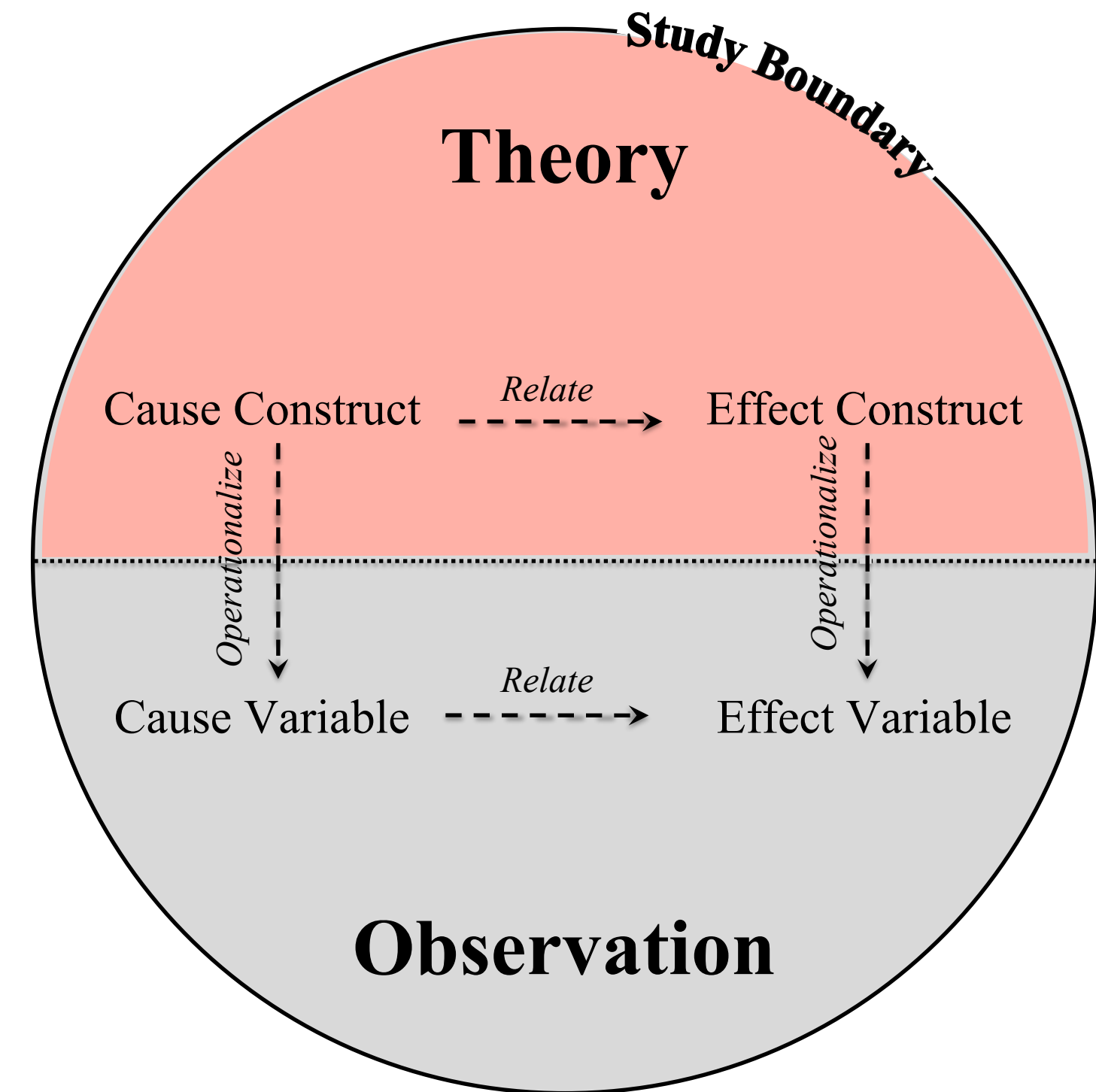
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VALIDITY AND RELIABILITY

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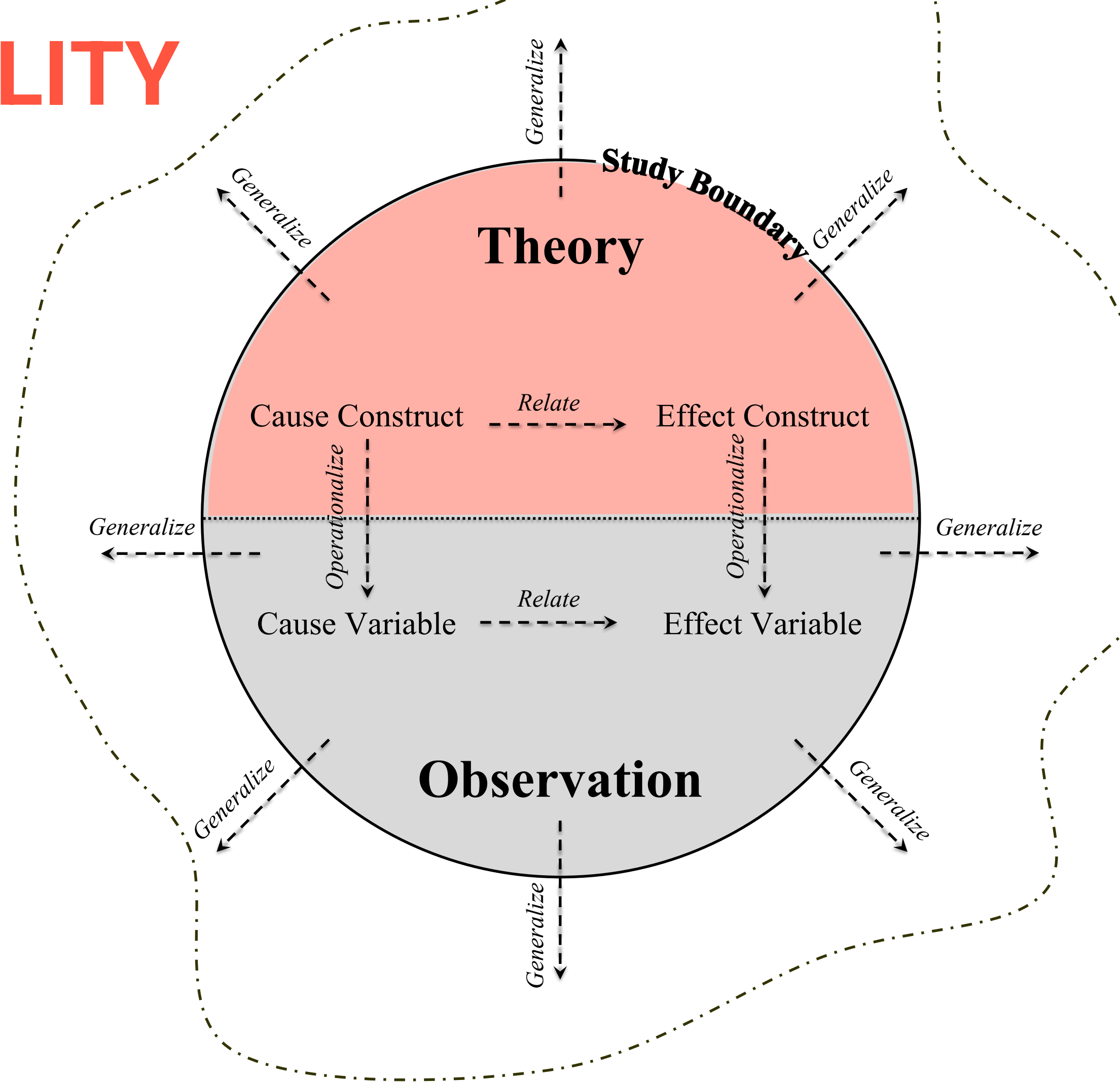
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VALIDITY AND RELIABILITY

Common dictionary

Text Analysis can be used to **operationalize** and **measure** management **constructs** by extracting structured indicators from unstructured textual data.



VALIDITY AND RELIABILITY

Objectives

1. **Using text to operationalise Innovation constructs:** transforming unstructured textual data into measurable variables that represent theoretical concepts central to innovation studies.
2. **Exploratory content analysis:** aims to uncover emerging themes, patterns, or narratives in textual datasets without predefined hypotheses. Often serving as an initial step before focused modeling or hypothesis testing, this approach supports the generation of new research questions and complements qualitative studies.
3. **Explanatory modelling:** This objective involves integrating text-based constructs into explanatory regression models. The goal is to isolate and test how specific text-based constructs relate to outcomes of theoretical interest. The researcher aims at testing causal hypotheses.
4. **Predicting innovation-related outcomes:** use of text analysis to forecast future or unknown events, outputs, or performance related to innovation process. This objective focuses on building models that use textual data as input to predict future or unobserved outcomes.

Berger, J., Humphreys, A., Ludwig, S., Moe, W. W., Netzer, O., & Schweidel, D. A. (2020). Uniting the tribes: Using text for marketing insight. Journal of marketing, 84(1), 1-25.

Herhausen, D., Ludwig, S., Abedin, E., Haque, N. U., & de Jong, D. (2025). From words to Insights: Text analysis in business research. Journal of Business Research, 198, 115491.

Feuerriegel, S., Maarouf, A., Bär, D., Geissler, D., Schweisthal, J., Pröllochs, N., ... & Van Bavel, J. J. (2025). Using natural language processing to analyse text data in behavioural science. Nature Reviews Psychology, 4(2), 96-111.

VALIDITY AND RELIABILITY

Validity & Reliability

Cronbach, L. J., & Meehl, P. E. (1955). Construct validity in psychological tests. *Psychological bulletin*, 52(4), 281.

Campbell, D. T. (1957). Factors relevant to the validity of experiments in social settings. *Psychological Bulletin*, 54(4), 297–312.

Campbell, D. T., & Fiske, D. W. (1959). Convergent and discriminant validation by the multitrait-multimethod matrix. *Psychological bulletin*, 56(2), 81.

Campbell, D. T., & Stanley, J. C. (1963). *Experimental and Quasi-Experimental Designs for Research*, Rand McNally & Company.

Cook, T. b., & Campbell, D. T. (1979). *Quasi-experimentation: Design and analysis issues for field settings*. Chicago: f-and-McNally.

Carmines, E. G., & Zeller, R. A. (1979). *Reliability and validity assessment*. Sage publications.

Kirk, J., & Miller, M. L. (1986). *Reliability and validity in qualitative research* (Vol. 1). Sage.

Nunnally, J. C., & Bernstein, I. H. (1994). *Psychometric theory* (3rd ed.). New York, NY: McGraw-Hill.

Shadish, W.R., Cook, T.D. and Campbell, D.T. (2002) *Experimental and Quasi-Experimental Designs for Generalized Causal Inference*. Houghton, Mifflin and Company.

Neuendorf, K. A. (2002). *The content analysis guidebook*. Thousand Oaks, CA: Sage.

DeVellis, R. F. (2012). *Scale development: Theory and applications* Sage publicationsSage Publications.

Scandura, T. A., & Williams, E. A. (2000). Research methodology in management: Current practices, trends, and implications for future research. *Academy of Management journal*, 43(6), 1248-1264.

Gill, J., & Johnson, P. (2010). *Research methods for managers*.

Weber R. P. (1990). *Basic content analysis* (2nd ed.). Thousand Oaks, CA: Sage.

Krippendorff, K. (2004). *Content analysis: An introduction to its methodology*. Thousand Oaks, CA: Sage.

Short, J. C., Broberg, J. C., Coglisier, C. C., & Brigham, K. H. (2010). Construct validation using computer-aided text analysis (CATA) an illustration using entrepreneurial orientation. *Organizational research methods*, 13(2), 320-347.

McKenny, A. F., Short, J. C., & Payne, G. T. (2013). Using computer-aided text analysis to elevate constructs: An illustration using psychological capital. *Organizational research methods*, 16(1), 152-184.

Berger, J., Humphreys, A., Ludwig, S., Moe, W. W., Netzer, O., & Schweidel, D. A. (2020). Uniting the tribes: Using text for marketing insight. *Journal of marketing*, 84(1), 1-25.

Campion, E. D., & Campion, M. A. (2025). A review of text analysis in human resource management research: Methodological diversity, constructs identified, and validation best practices. *Human Resource Management Review*, 35(2), 101078.

Pandey, S., & Pandey, S. K. (2019). Applying natural language processing capabilities in computerized textual analysis to measure organizational culture. *Organizational Research Methods*, 22(3), 765-797.

VALIDITY AND RELIABILITY

Validity & Reliability

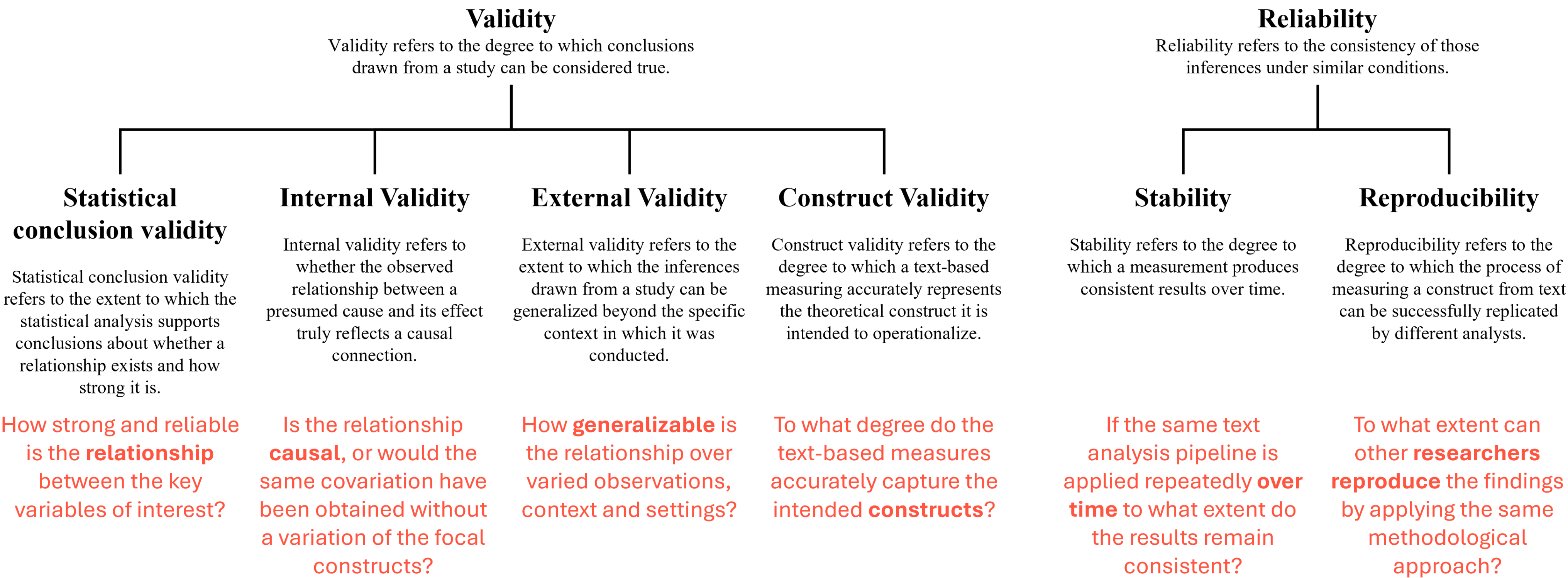
More than **30 different dimensions** of validity and reliability.

No agreement on the definitions



VALIDITY AND RELIABILITY

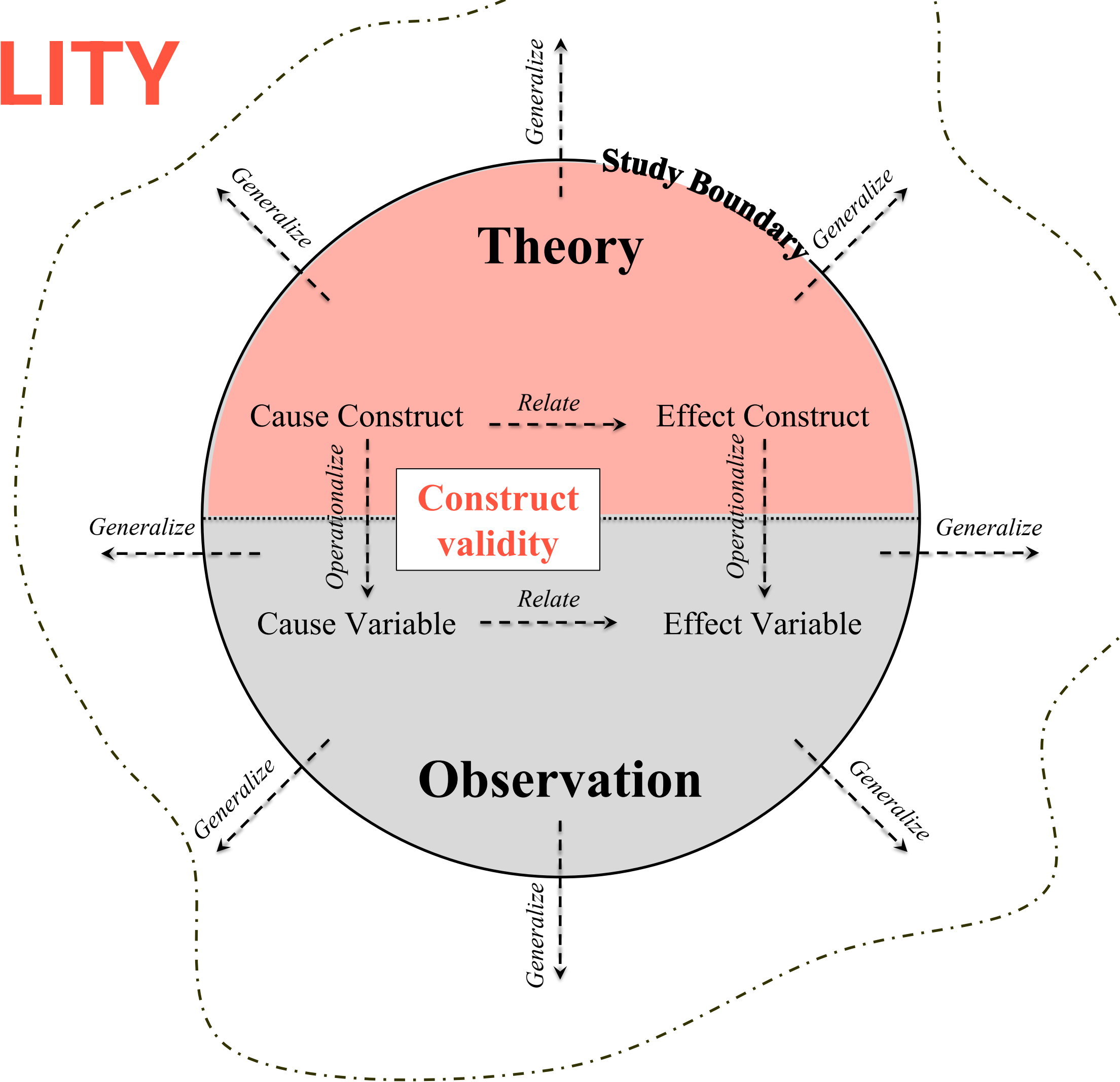
Validity & Reliability



VALIDITY AND RELIABILITY

Rigor Dimensions

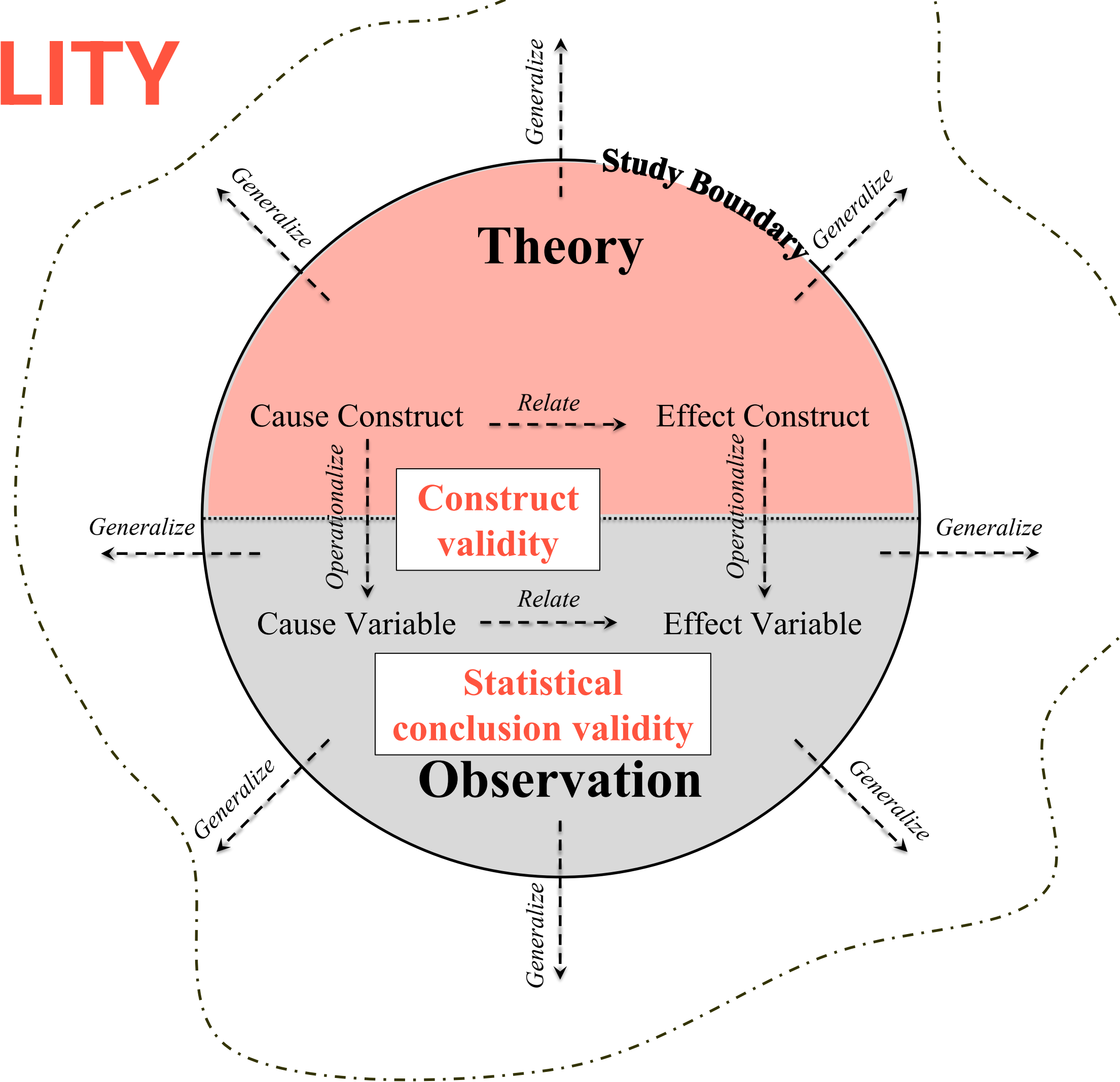
Understanding the **different dimensions** of validity requires clarifying where they arise within the research process.



VALIDITY AND RELIABILITY

Rigor Dimensions

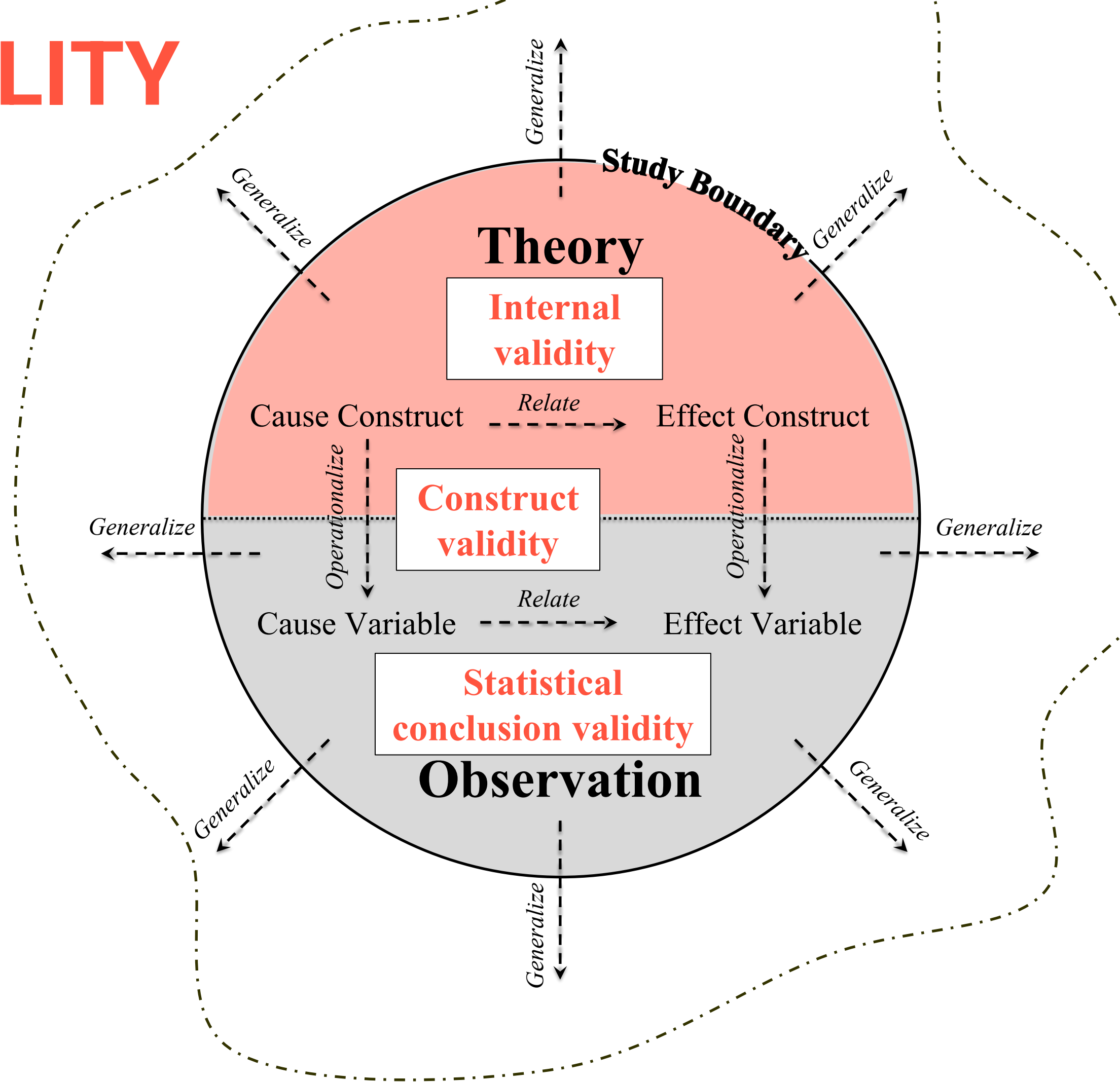
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VALIDITY AND RELIABILITY

Rigor Dimensions

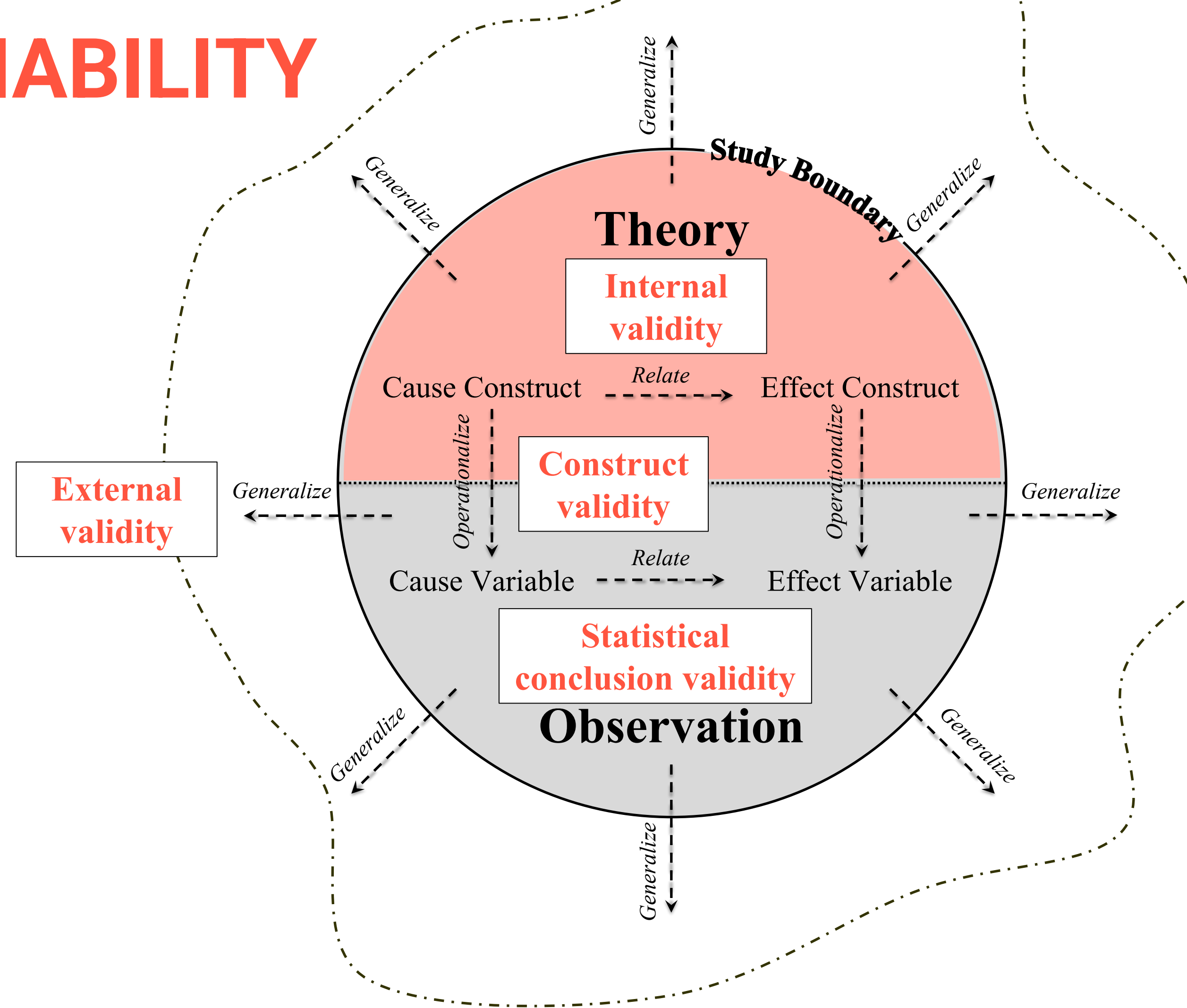
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VALIDITY AND RELIABILITY

Rigor Dimensions

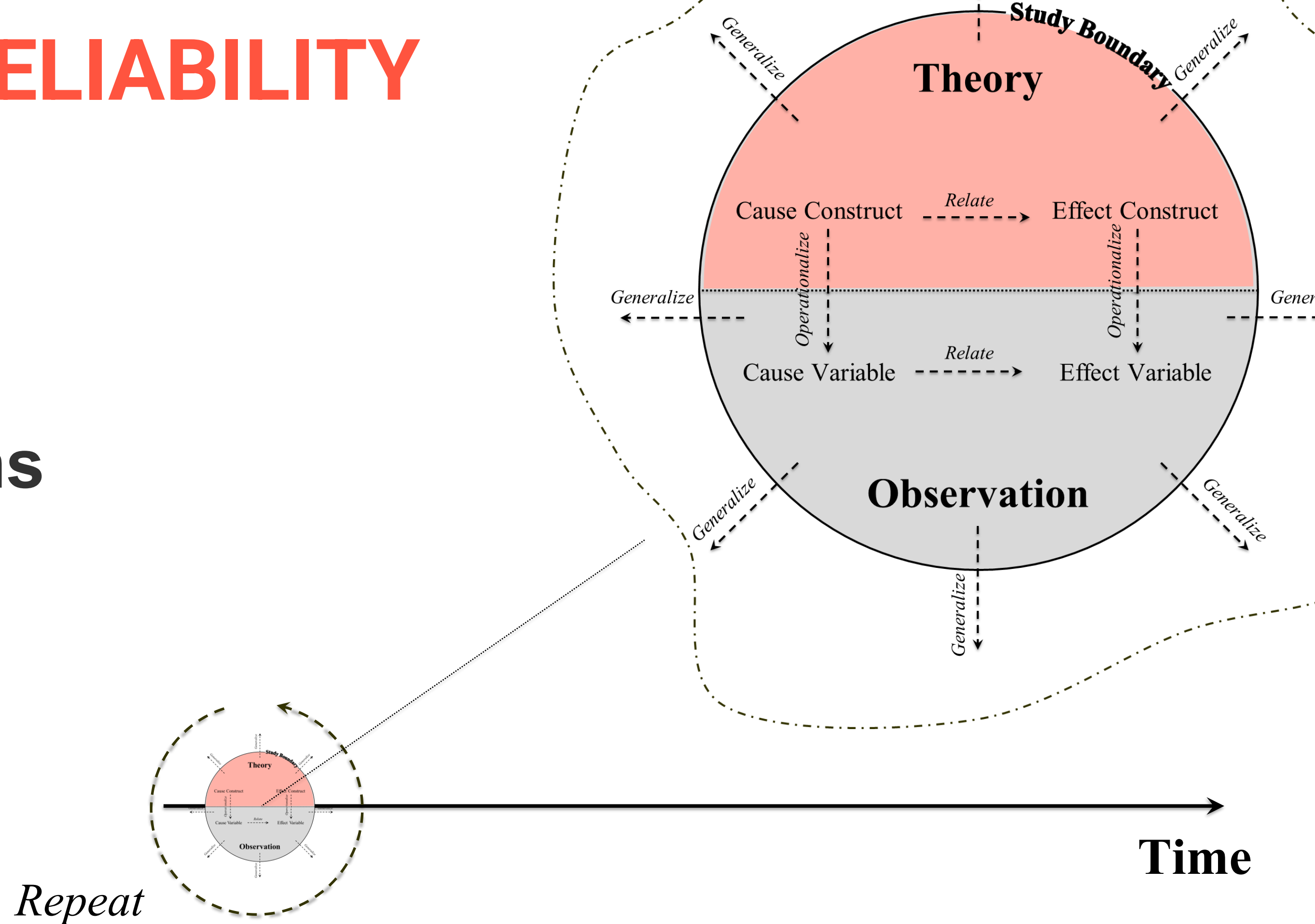
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VALIDITY AND RELIABILITY

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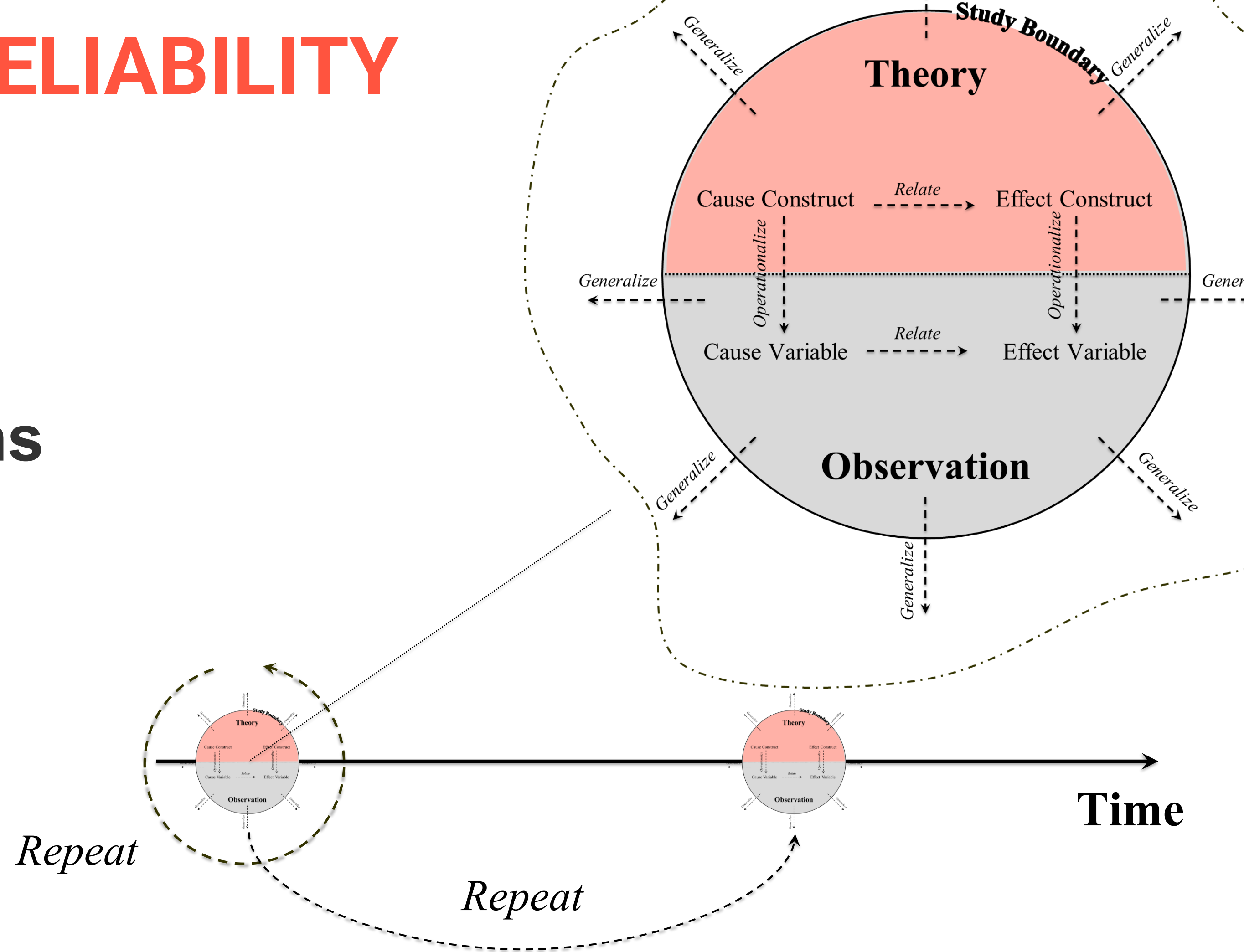
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VALIDITY AND RELIABILITY

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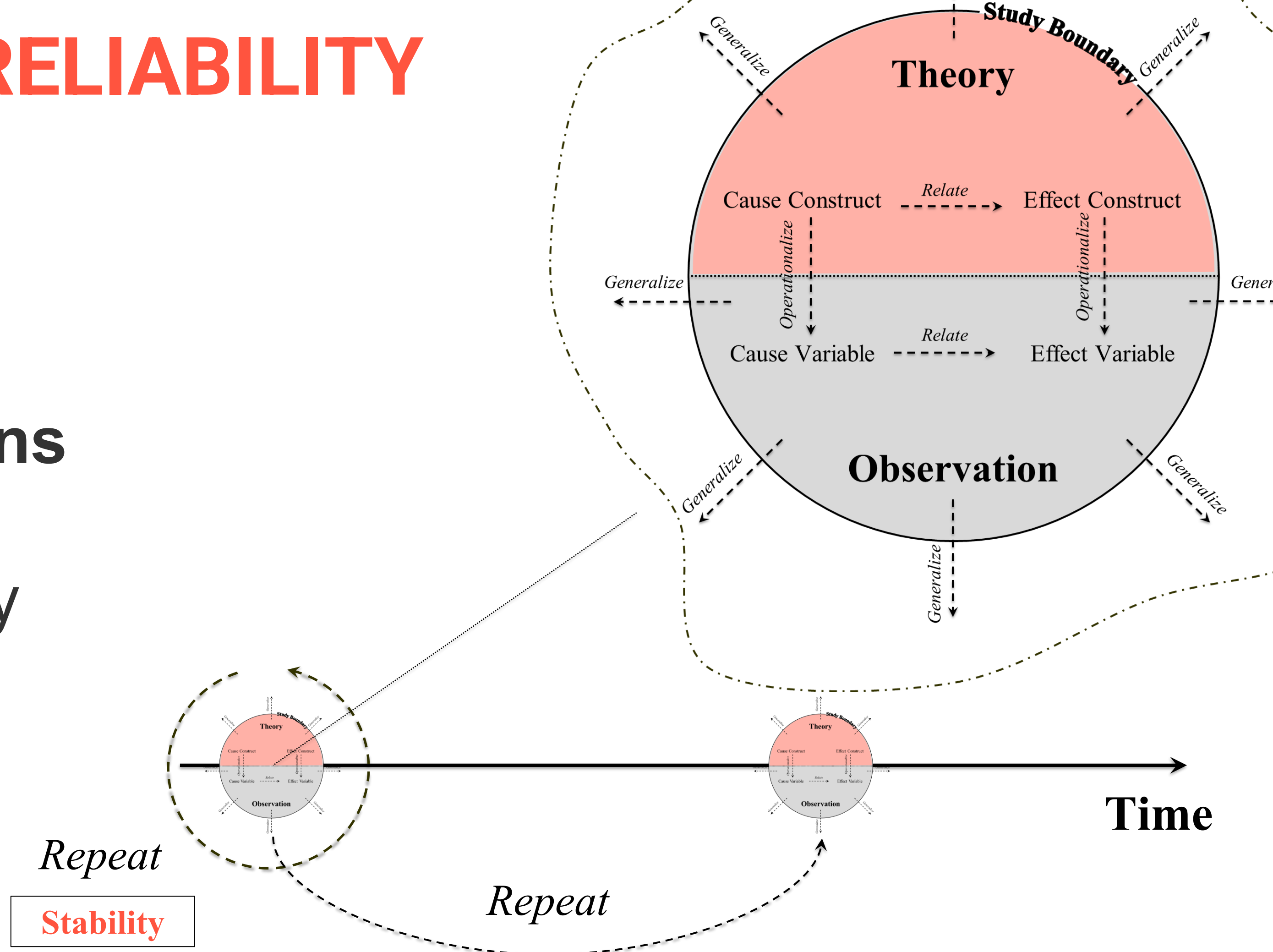
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VALIDITY AND RELIABILITY

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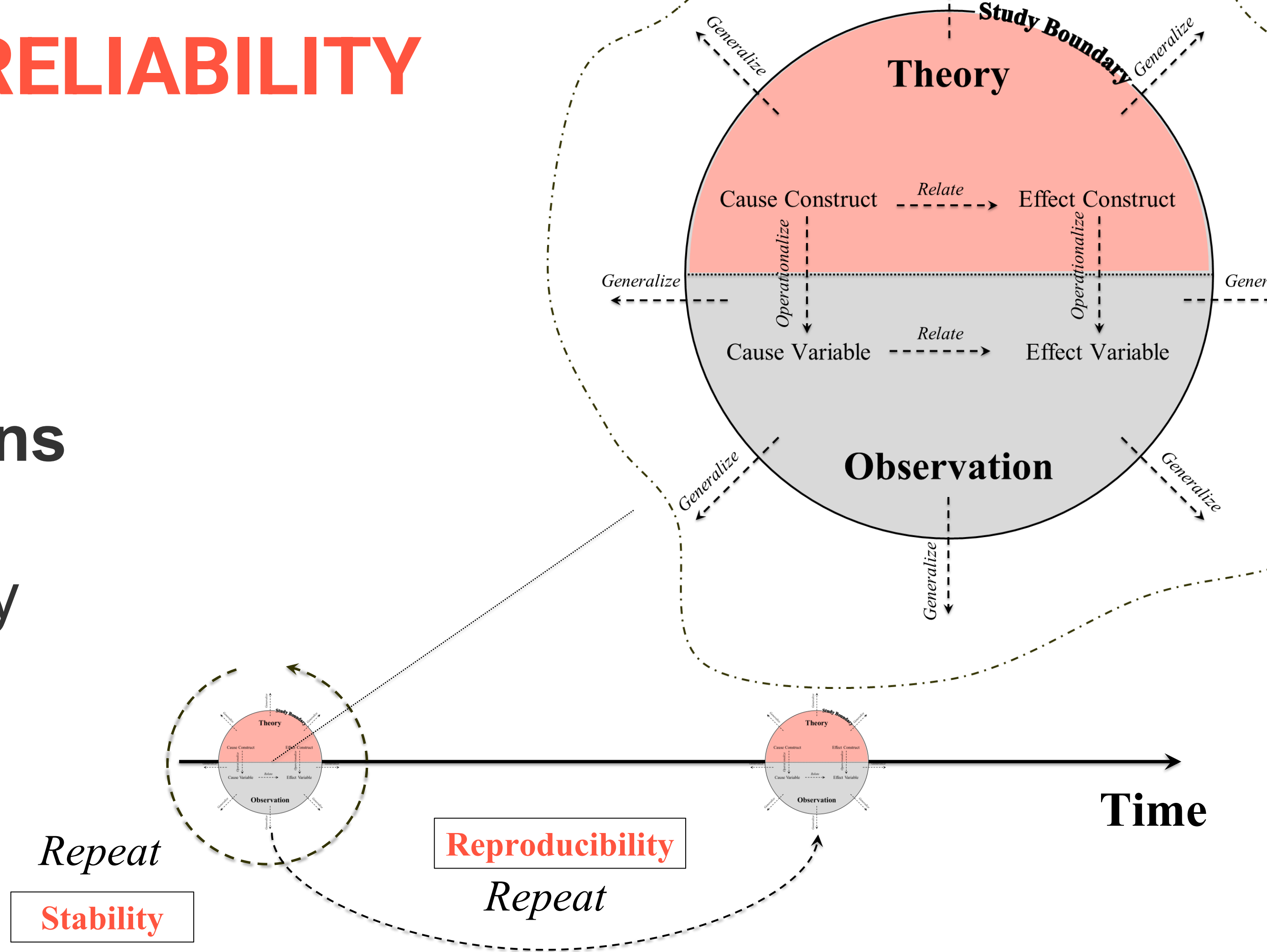
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VALIDITY AND RELIABILITY

Rigor Dimensions

Understanding the **different dimensions** of validity requires clarifying where they arise within the research process.



VALIDITY AND RELIABILITY

Characteristics of Text Analysis

Understand the **specific characteristics of text analysis** as research method

2



VALIDITY AND RELIABILITY

Characteristics of Text Analysis

Understand the **specific characteristics of text analysis** as research method

2



Statement 1

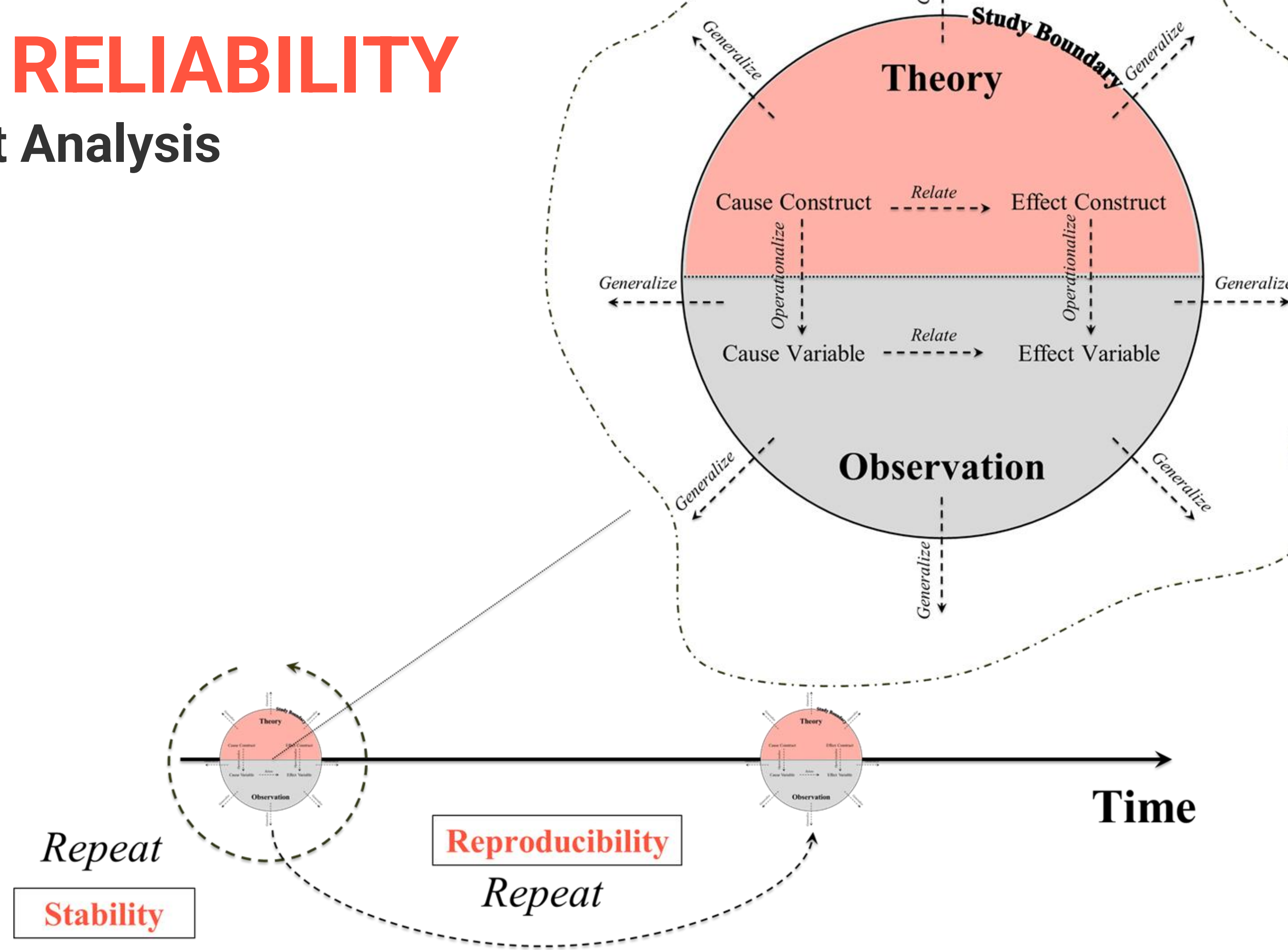
*“Text analysis is typically used as a **non-experimental** research method. In fact, it is typically applied to **secondary data**.”*

Statement 2

*“Text analysis is **computational method**, and like all methods it relies on multiple assumptions. Without a clear understanding of these assumptions, researchers may produce invalid results.”*

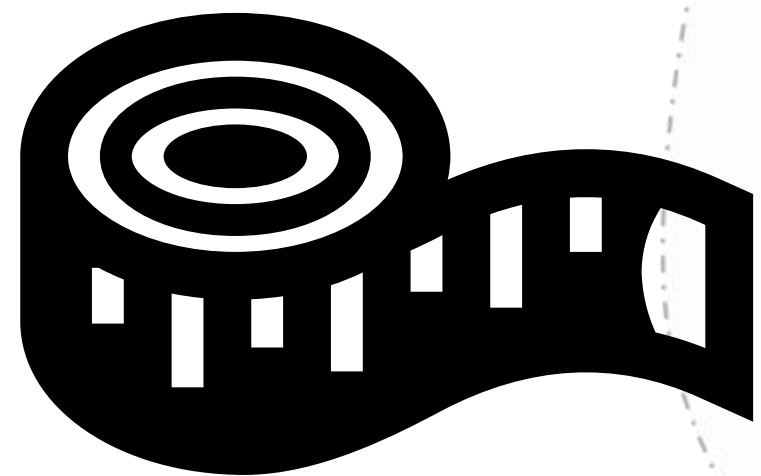
VALIDITY AND RELIABILITY

Characteristics of Text Analysis

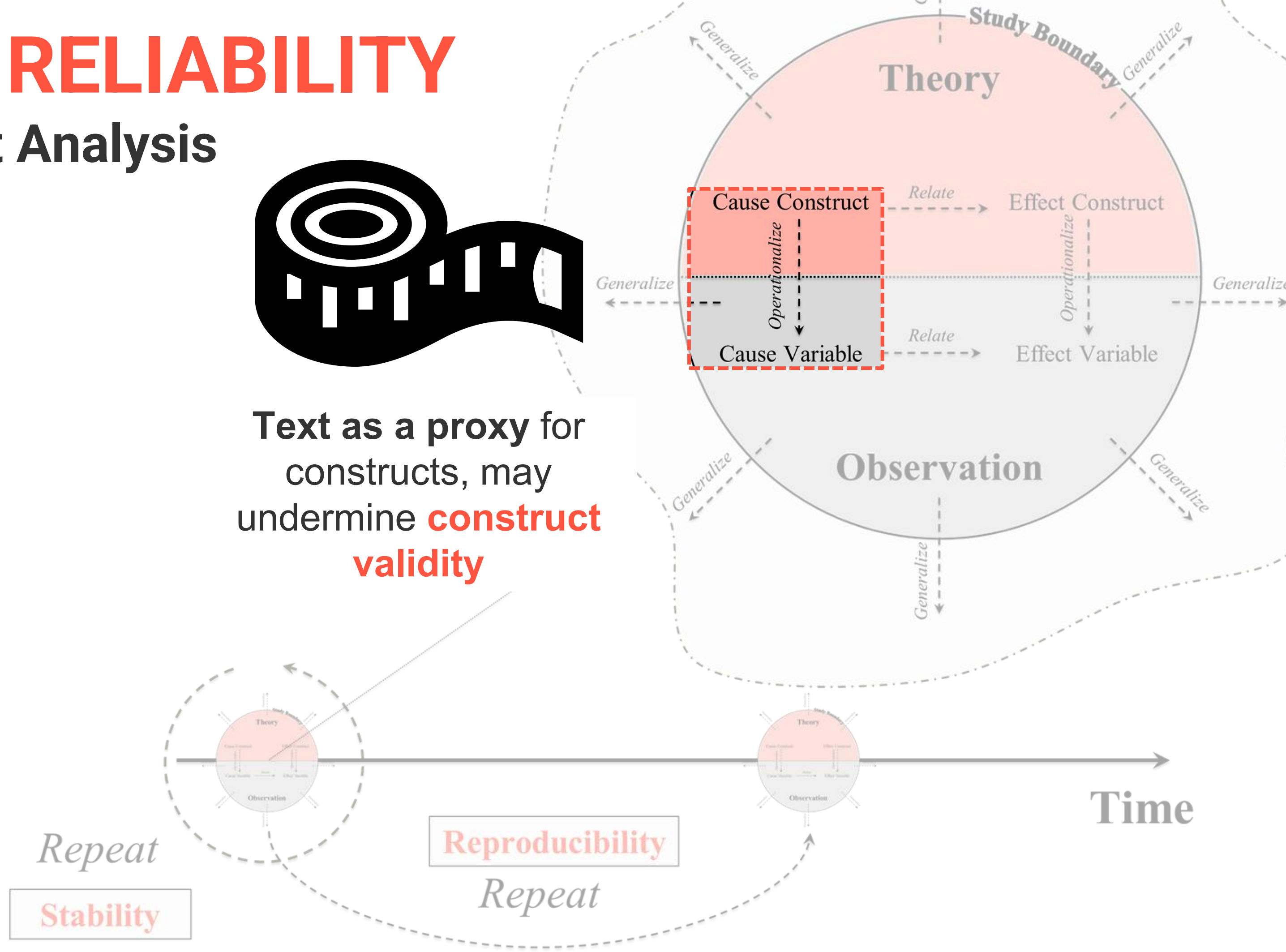


VALIDITY AND RELIABILITY

Characteristics of Text Analysis

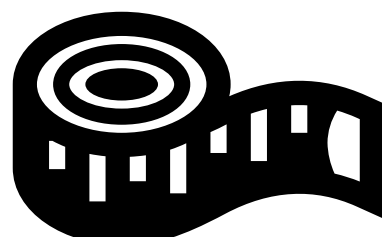


Text as a proxy for constructs, may undermine **construct validity**



VALIDITY AND RELIABILITY

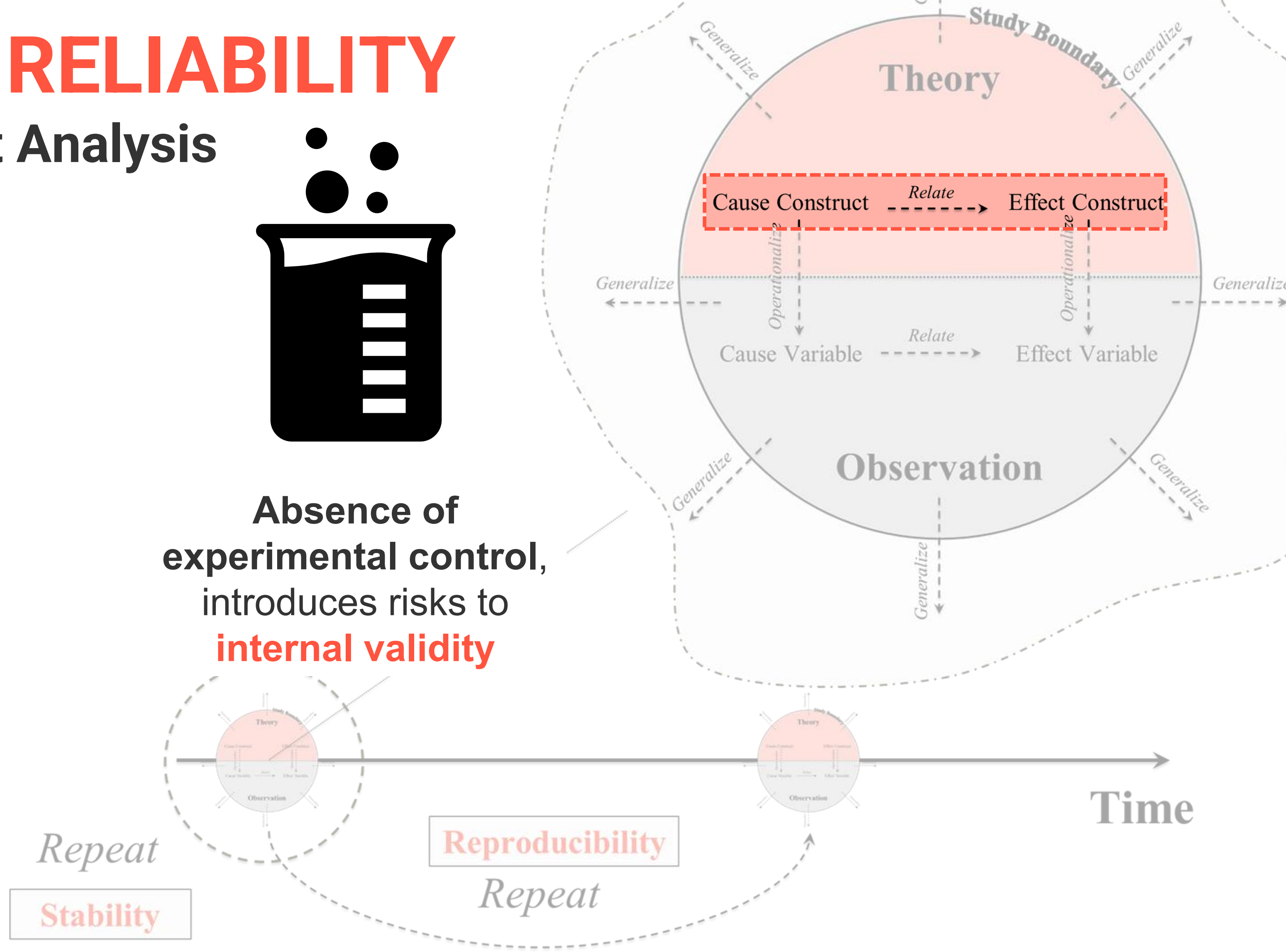
Characteristics of Text Analysis



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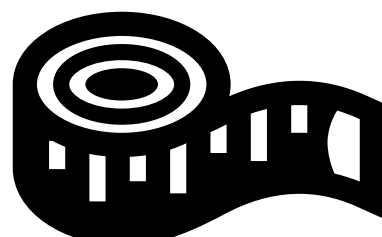


Absence of experimental control, introduces risks to **internal validity**

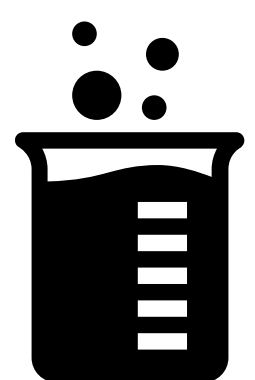


VALIDITY AND RELIABILITY

Characteristics of Text Analysis



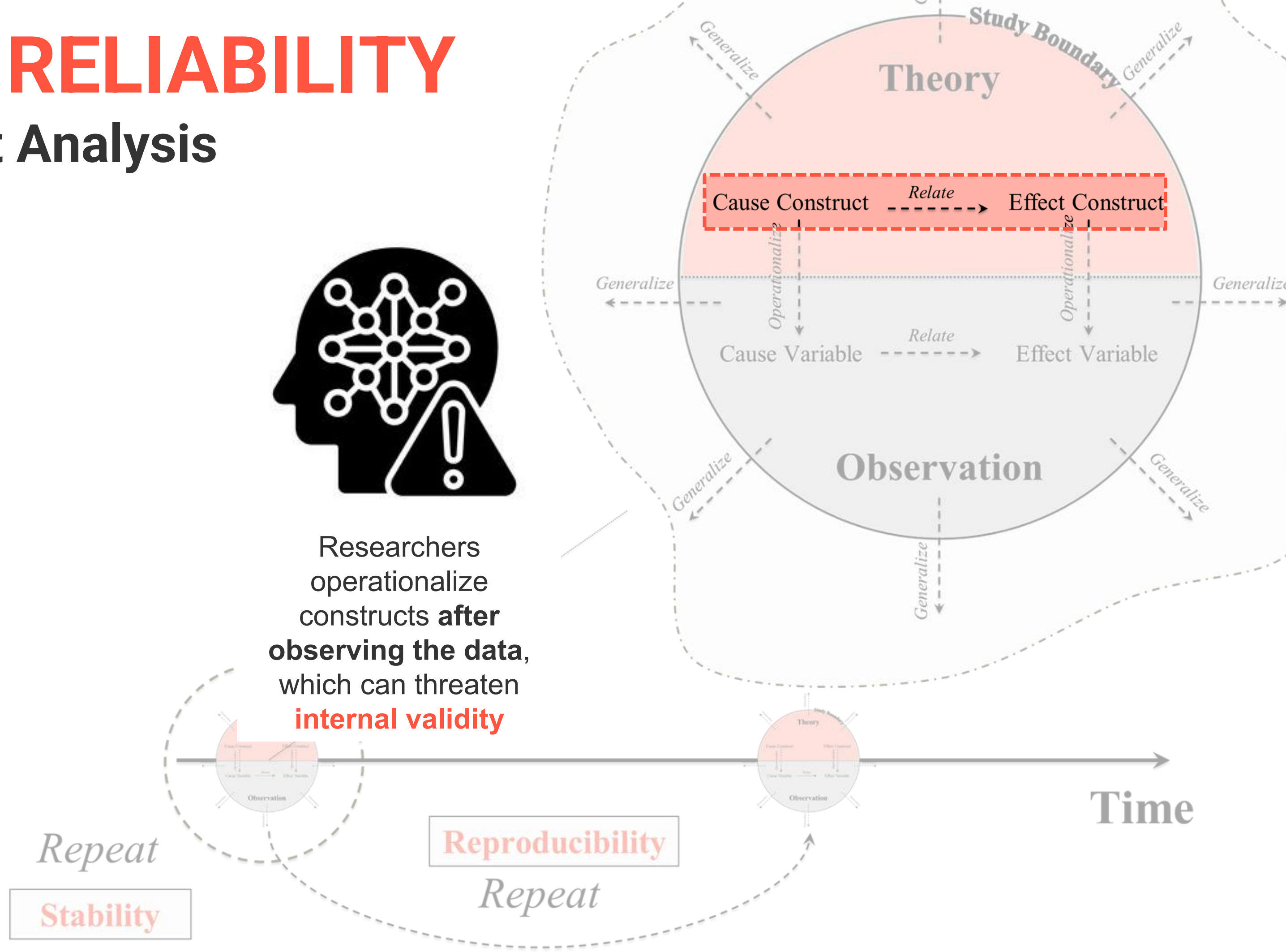
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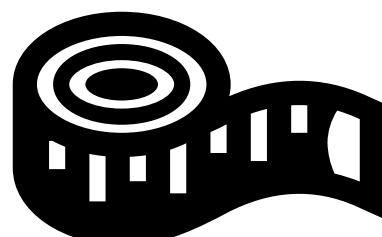


Researchers operationalize constructs **after** observing the data, which can threaten **internal validity**

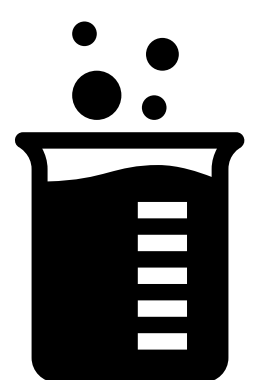


VALIDITY AND RELIABILITY

Characteristics of Text Analysis



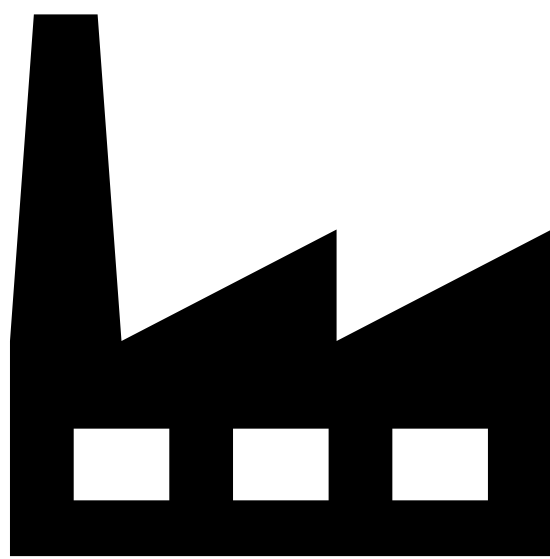
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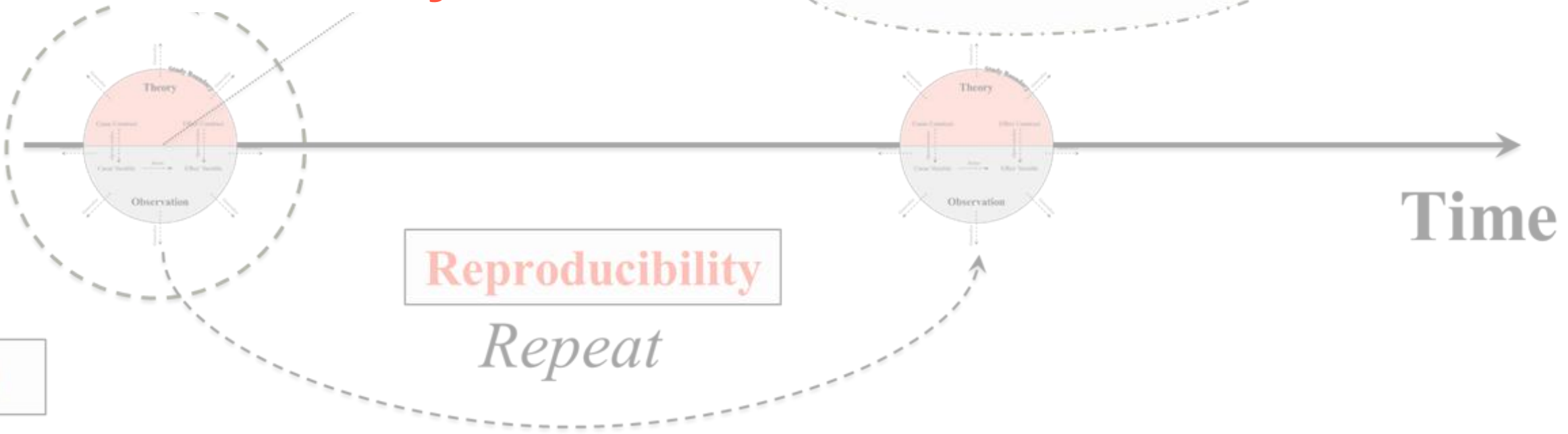
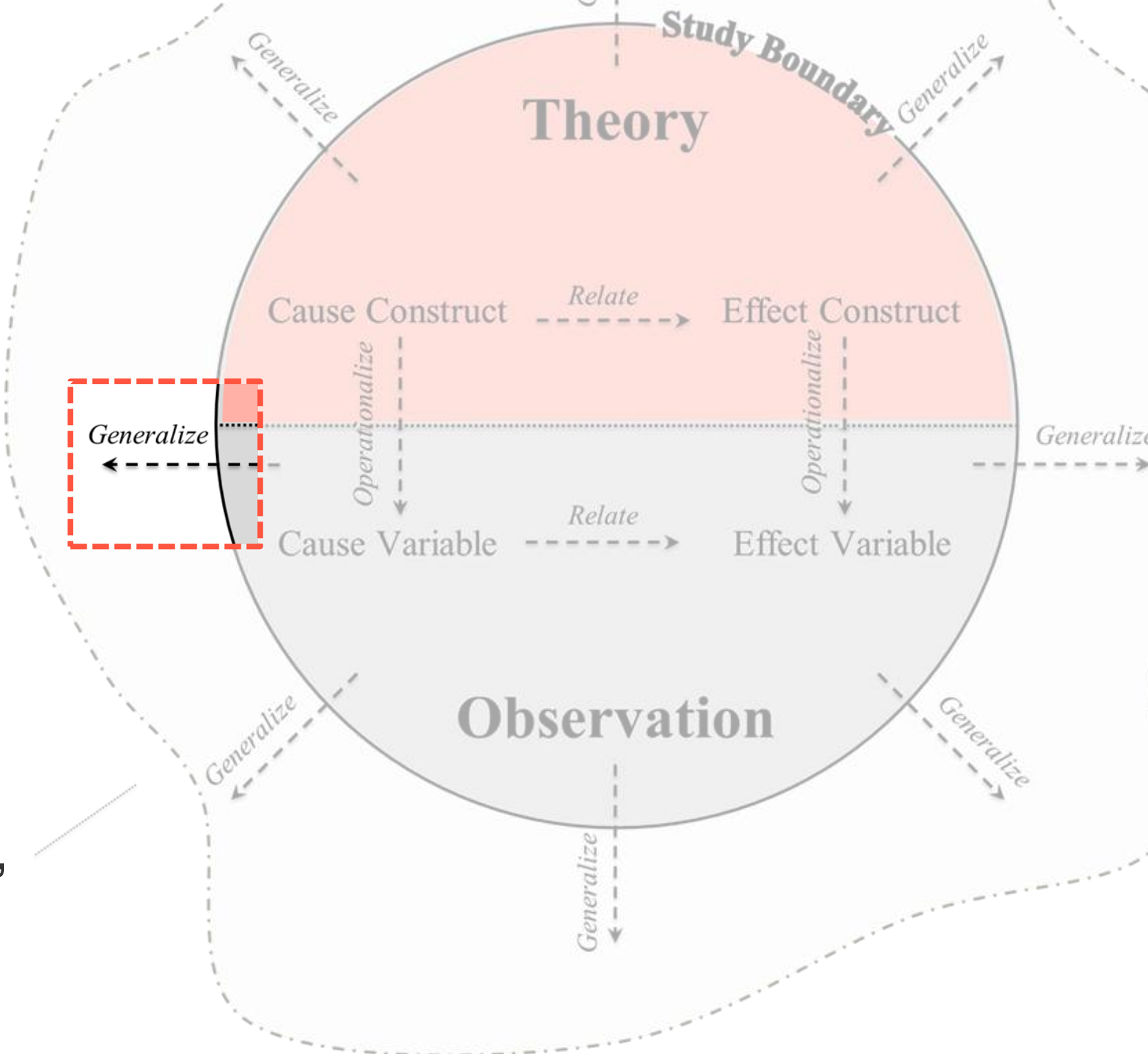
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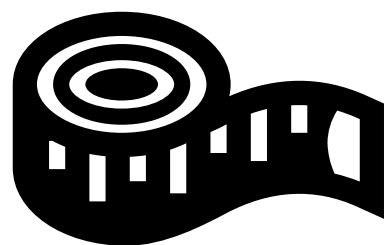


Real-world data, higher **external validity**

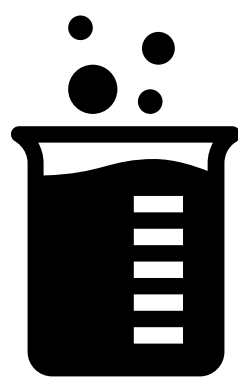


VALIDITY AND RELIABILITY

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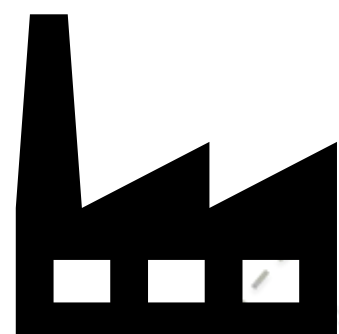
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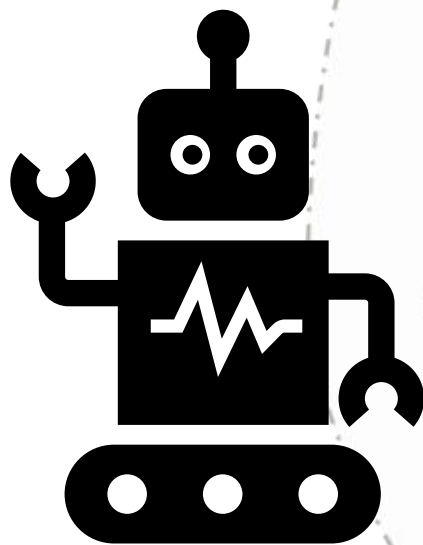
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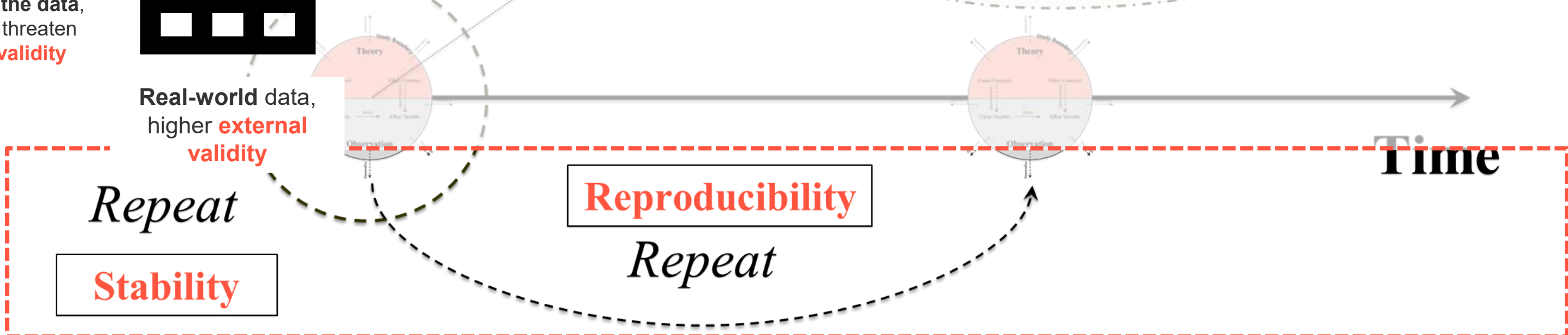
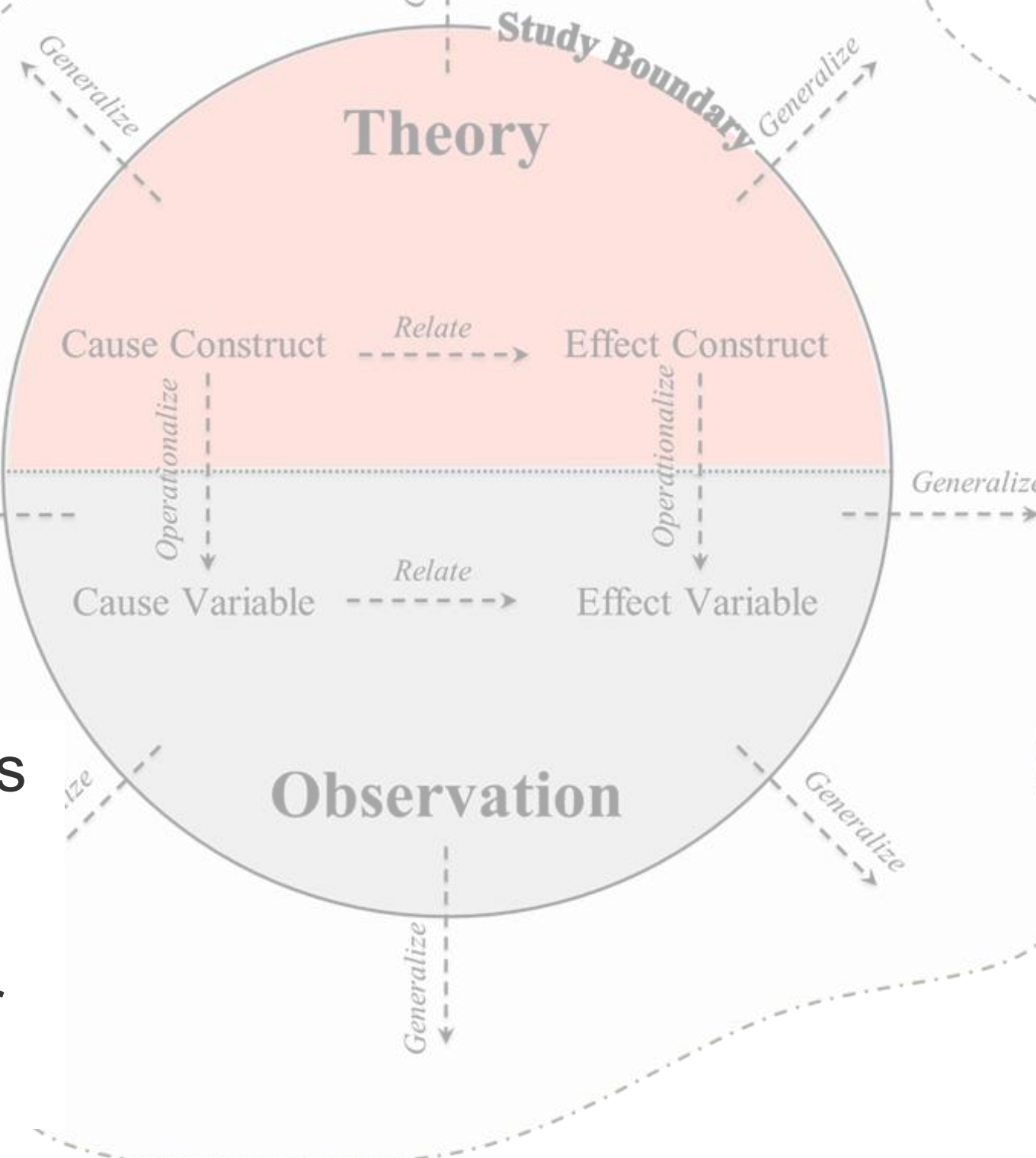
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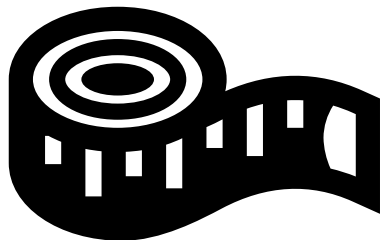


Compared to methods that rely heavily on **human judgment**, LLMs exhibits higher **reliability**



VALIDITY AND RELIABILITY

Characteristics of Text Analysis



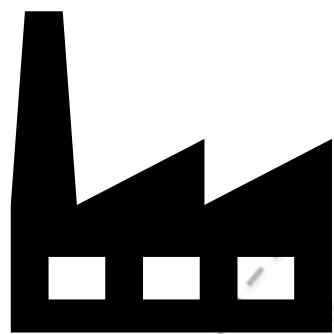
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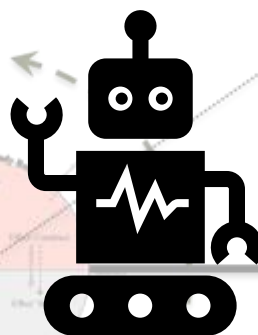
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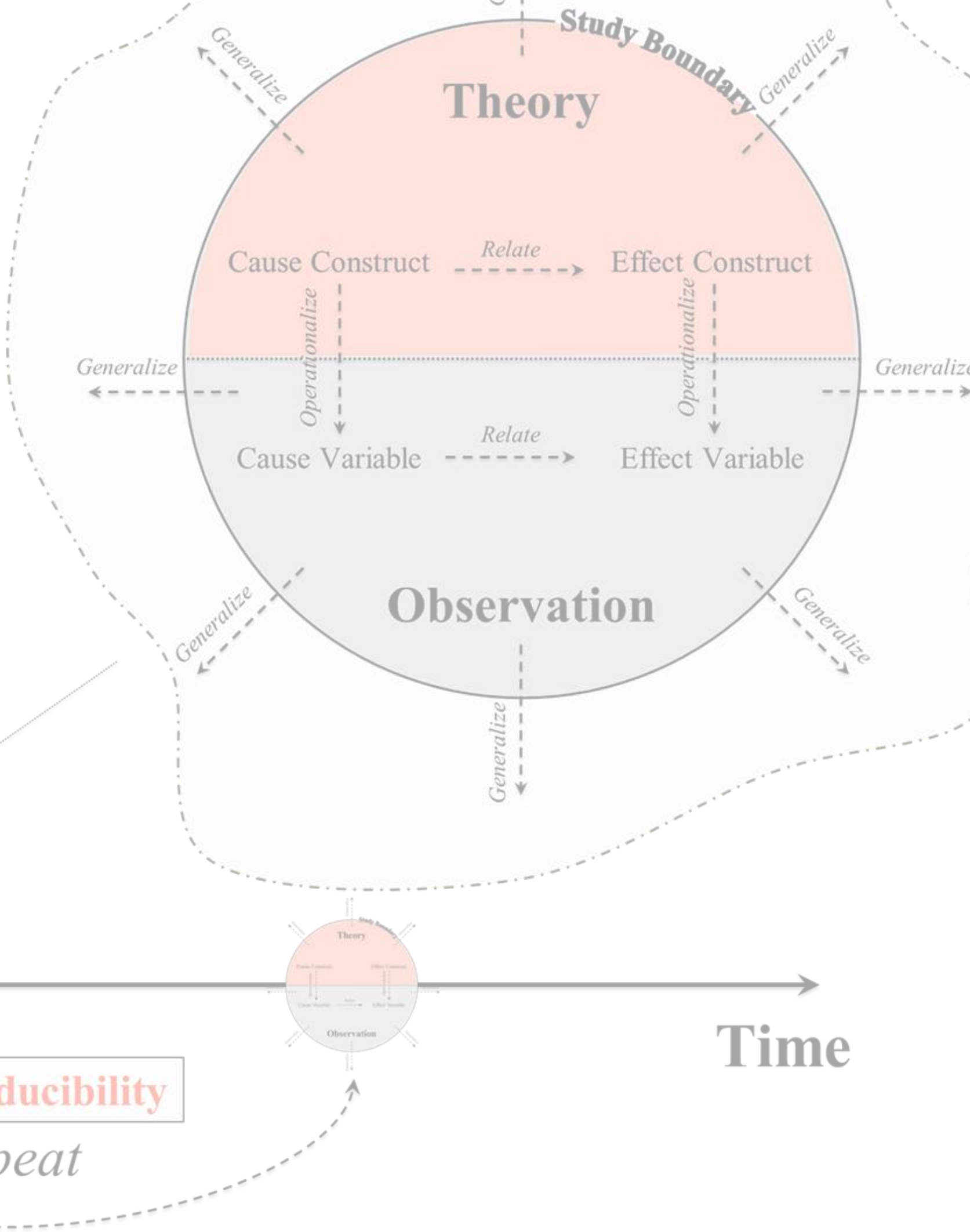
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Repeat

Stability

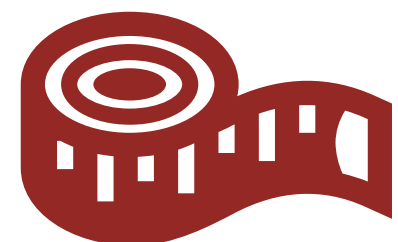
Reproducibility

Repeat



VALIDITY AND RELIABILITY

Characteristics of Text Analysis



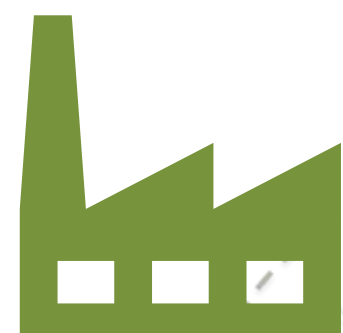
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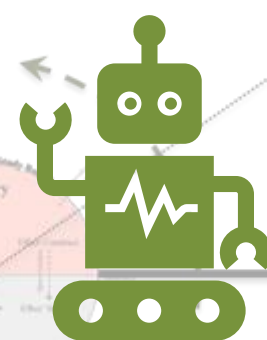


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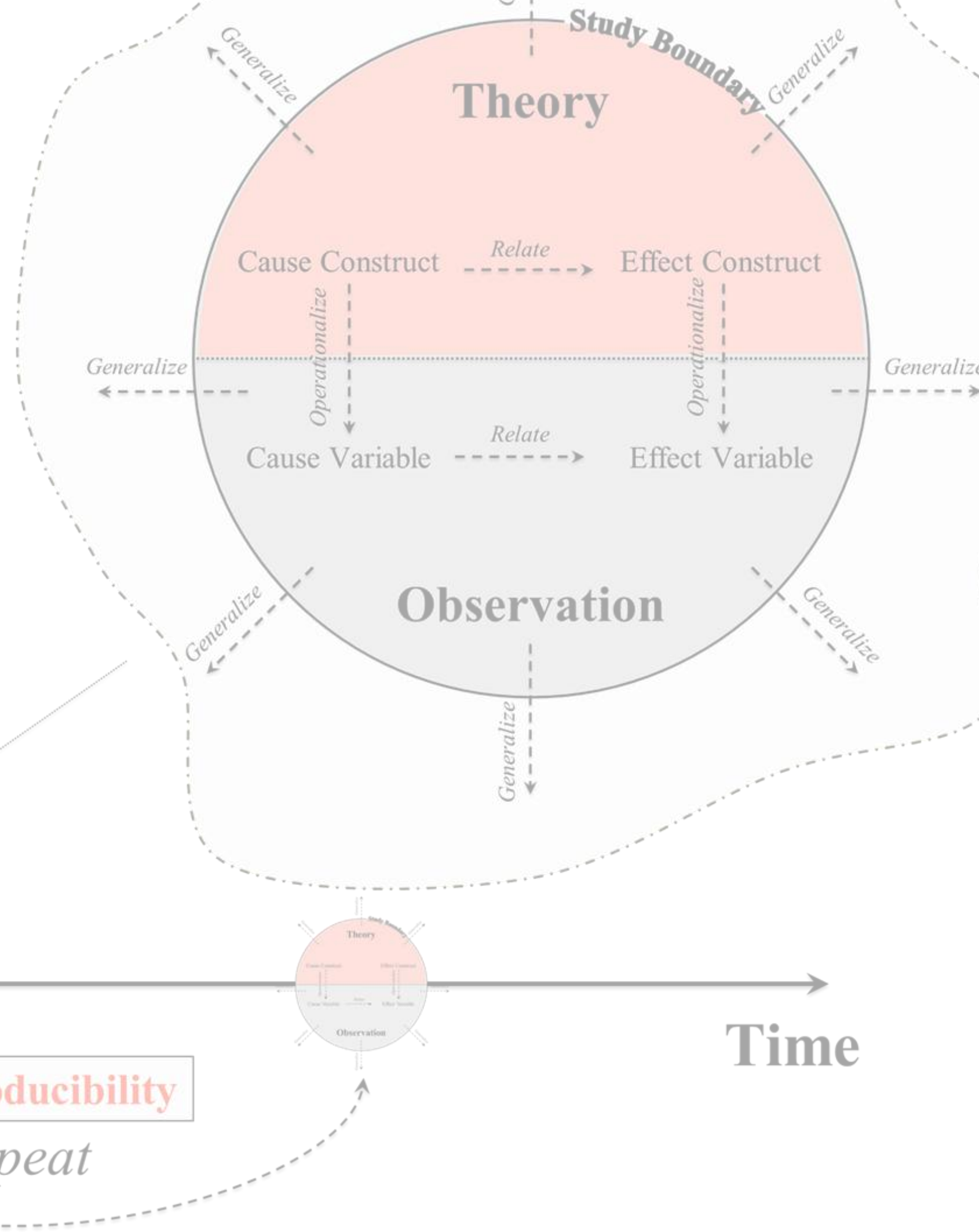
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04 RIGOR IN MANAGEMENT

RIGOR IN MANAGEMENT

Practices

Examine how rigor strategies are applied in management using text analysis

3



RIGOR IN MANAGEMENT

Practices

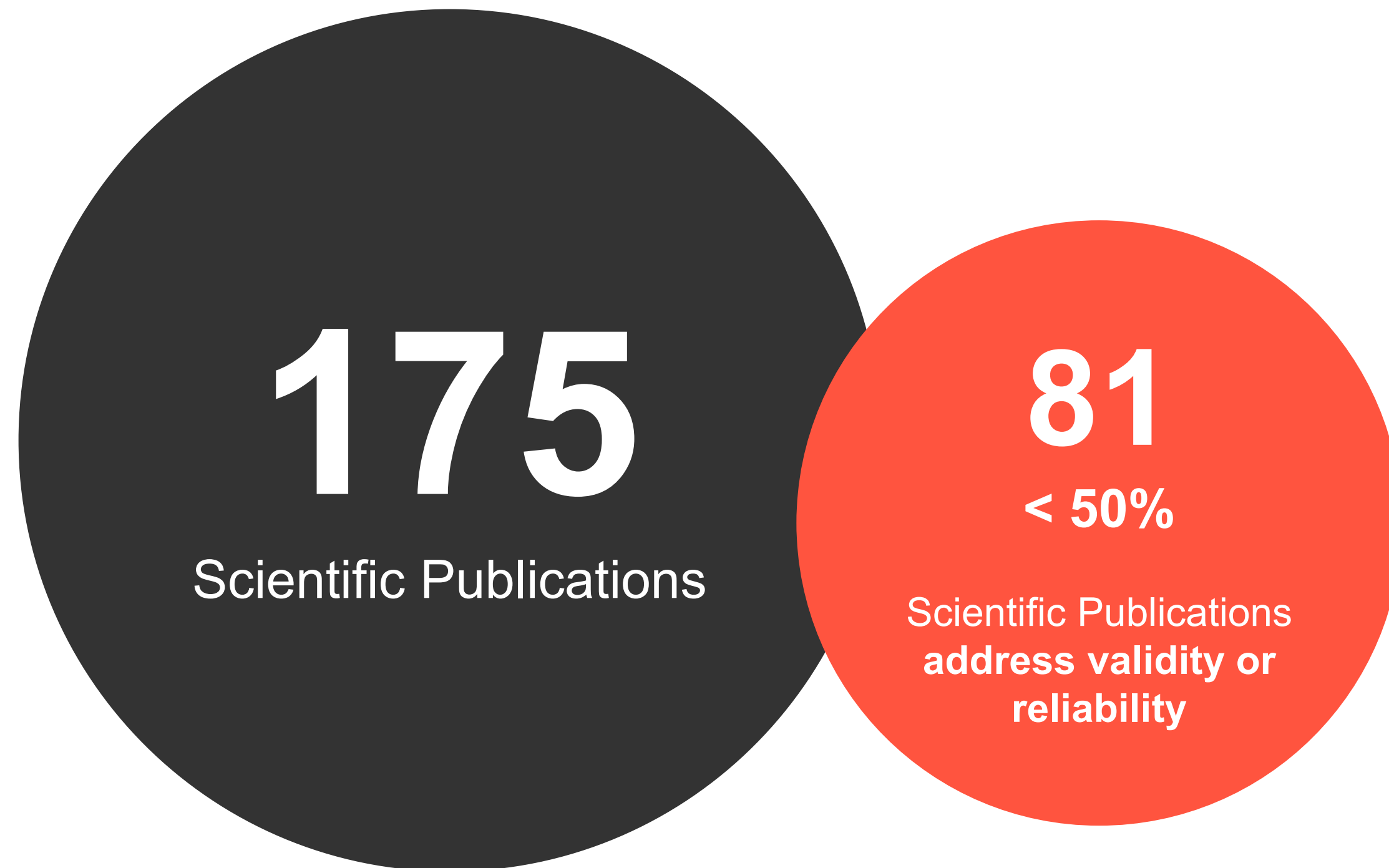
175

Scientific Publications

Scientific publications on
Innovation Management

RIGOR IN MANAGEMENT

Practices



Out of the 175 articles analyzed, only **81 (less than 50%)** address issues related to validity or reliability.

RIGOR IN MANAGEMENT

Practices - Primary Report

In the primary report I analysed whether the publication **explicitly mentions** at least one form of validity or reliability.

Journal	Total No. of Papers	Validity	Reliability	Validity type				Reliability type	
				Statistical Conclusion Validity	Internal Validity	External Validity	Construct Validity	Stability	Reproducibility
Academy of Management Journal (4*)	2	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)
Management Science (4*)	3	3 (100.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	2 (66.67%)	0 (0.00%)	0 (0.00%)
Organization Science (4*)	1	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)
Strategic Management Journal (4*)	5	4 (80.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	3 (60.00%)	2 (40.00%)	0 (0.00%)	0 (0.00%)
Research Policy (4*)	16	9 (56.25%)	1 (6.25%)	0 (0.00%)	1 (6.25%)	3 (18.75%)	3 (18.75%)	0 (0.00%)	1 (6.25%)
Journal of Product Innovation Management (4)	6	3 (50.00%)	0 (0.00%)	0 (0.00%)	1 (16.67%)	1 (16.67%)	2 (33.34%)	0 (0.00%)	0 (0.00%)
Industry and Innovation (3)	2	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)
Journal of Technology Transfer (3)	8	2 (25.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	1 (12.50%)	0 (0.00%)	0 (0.00%)	1 (12.50%)
R and D Management (3)	8	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)
Technological Forecasting and Social Change (3)	109	36 (33.03%)	3 (2.75%)	0 (0.00%)	0 (0.00%)	3 (2.75%)	8 (7.34%)	0 (0.00%)	3 (2.75%)
Technovation (3)	15	6 (40.00%)	0 (0.00%)	1 (6.67%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)
Total	175	63 (36.00%)	4 (2.29%)	1 (0.57%)	2 (1.14%)	11 (6.29%)	17 (9.71%)	0 (0.00%)	4 (2.29%)

RIGOR IN MANAGEMENT

Practices - Secondary Report

In the secondary report I analysed whether the publication **addresses at least one threat to validity or reliability**, even if not explicitly labeled as such.

Journal	Total No. of Papers	Validity	Reliability	Validity type				Reliability type	
				Statistical Conclusion Validity	Internal Validity	External Validity	Construct Validity	Stability	Reproducibility
Academy of Management Journal (4*)	2	2 (100.00%)	0 (0.00%)	1 (50.00%)	2 (100.00%)	1 (50.00%)	2 (100.00%)	0 (0.00%)	0 (0.00%)
Management Science (4*)	3	3 (100.00%)	3 (100.00%)	1 (33.33%)	1 (33.33%)	1 (33.33%)	3 (100.00%)	0 (0.00%)	3 (100.00%)
Organization Science (4*)	1	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)
Strategic Management Journal (4*)	5	5 (100.00%)	3 (60.00%)	2 (40.00%)	4 (80.00%)	4 (80.00%)	5 (100.00%)	2 (40.00%)	2 (40.00%)
Research Policy (4*)	16	12 (75.00%)	3 (18.75%)	3 (18.75%)	7 (43.75%)	8 (50.00%)	12 (75.00%)	0 (0.00%)	3 (18.75%)
Journal of Product Innovation Management (4)	6	4 (66.67%)	0 (0.00%)	2 (33.33%)	1 (16.67%)	3 (50.00%)	4 (66.67%)	0 (0.00%)	0 (0.00%)
Industry and Innovation (3)	2	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)
Journal of Technology Transfer (3)	8	4 (50.00%)	1 (12.50%)	2 (25.00%)	2 (25.00%)	0 (0.00%)	3 (37.50%)	0 (0.00%)	1 (12.50%)
R and D Management (3)	8	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)
Technological Forecasting and Social Change (3)	109	42 (38.53%)	10 (9.17%)	11 (10.09%)	11 (10.09%)	22 (20.18%)	37 (33.94%)	0 (0.00%)	10 (9.17%)
Technovation (3)	15	6 (40.00%)	3 (20.00%)	4 (26.67%)	2 (13.33%)	2 (13.33%)	6 (40.00%)	2 (13.33%)	1 (6.67%)
Total	175	78 (44.57%)	23 (13.14%)	26 (14.86%)	30 (17.14%)	41 (23.43%)	72 (41.14%)	4 (2.29%)	20 (11.43%)

RIGOR IN MANAGEMENT

Priorities - Context

Objectives	Total No. of Papers	Validity	Reliability	Validity type				Reliability type	
				Statistical Conclusion Validity	Internal Validity	External Validity	Construct Validity	Stability	Reproducibility
Operationalising innovation constructs	14	14 (100.00%)	9 (64.29%)	3 (21.43%)	3 (21.43%)	9 (64.29%)	14 (100.00%)	1 (7.14%)	9 (64.29%)
Exploratory content analysis	108	24 (22.22%)	7 (6.48%)	5 (4.63%)	4 (3.70%)	9 (8.33%)	21 (19.44%)	1 (0.93%)	6 (5.56%)
Explanatory modelling	35	27 (77.14%)	2 (5.71%)	14 (40.00%)	23 (65.71%)	15 (42.86%)	24 (68.57%)	1 (2.86%)	1 (2.86%)
Predicting outcomes	18	13 (72.22%)	5 (27.78%)	4 (22.22%)	0 (0.00%)	8 (44.44%)	13 (72.22%)	1 (5.56%)	4 (22.22%)

RIGOR IN MANAGEMENT

Practices - Context

Text Modelling Approach	Total No. of Papers	Validity	Reliability	Validity type				Reliability type	
				Statistical Conclusion Validity	Internal Validity	External Validity	Construct Validity	Stability	Reproducibility
Word count	15	2 (13.33%)	1 (6.67%)	0 (0.00%)	0 (0.00%)	1 (6.67%)	2 (13.33%)	0 (0.00%)	1 (6.67%)
Dictionary-based approach	65	31 (47.69%)	6 (9.23%)	16 (24.62%)	21 (32.31%)	17 (26.15%)	22 (33.85%)	0 (0.00%)	6 (9.23%)
Representation-based machine learning	23	21 (91.30%)	7 (30.43%)	6 (26.09%)	4 (17.39%)	13 (56.52%)	21 (91.30%)	3 (13.04%)	5 (21.74%)
Large Language Model	1	1 (100.00%)	0 (0.00%)	1 (100.00%)	0 (0.00%)	0 (0.00%)	1 (100.00%)	0 (0.00%)	0 (0.00%)
Topic modelling	54	16 (29.63%)	3 (5.56%)	1 (1.85%)	7 (12.96%)	9 (16.67%)	13 (24.07%)	1 (1.85%)	2 (3.70%)
Text similarity	41	21 (51.22%)	9 (21.95%)	4 (9.76%)	4 (9.76%)	13 (31.71%)	19 (46.34%)	0 (0.00%)	9 (21.95%)
Text clustering	10	2 (20.00%)	1 (10.00%)	1 (10.00%)	0 (0.00%)	0 (0.00%)	2 (20.00%)	0 (0.00%)	1 (10.00%)

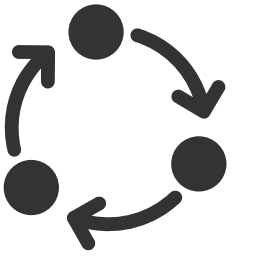
In general, studies employing **machine learning approaches** are more likely to examine construct validity and to incorporate more extensive assessments of validity.

RIGOR IN MANAGEMENT

Guidelines

Develop ad hoc methods to ensure rigor in text analysis

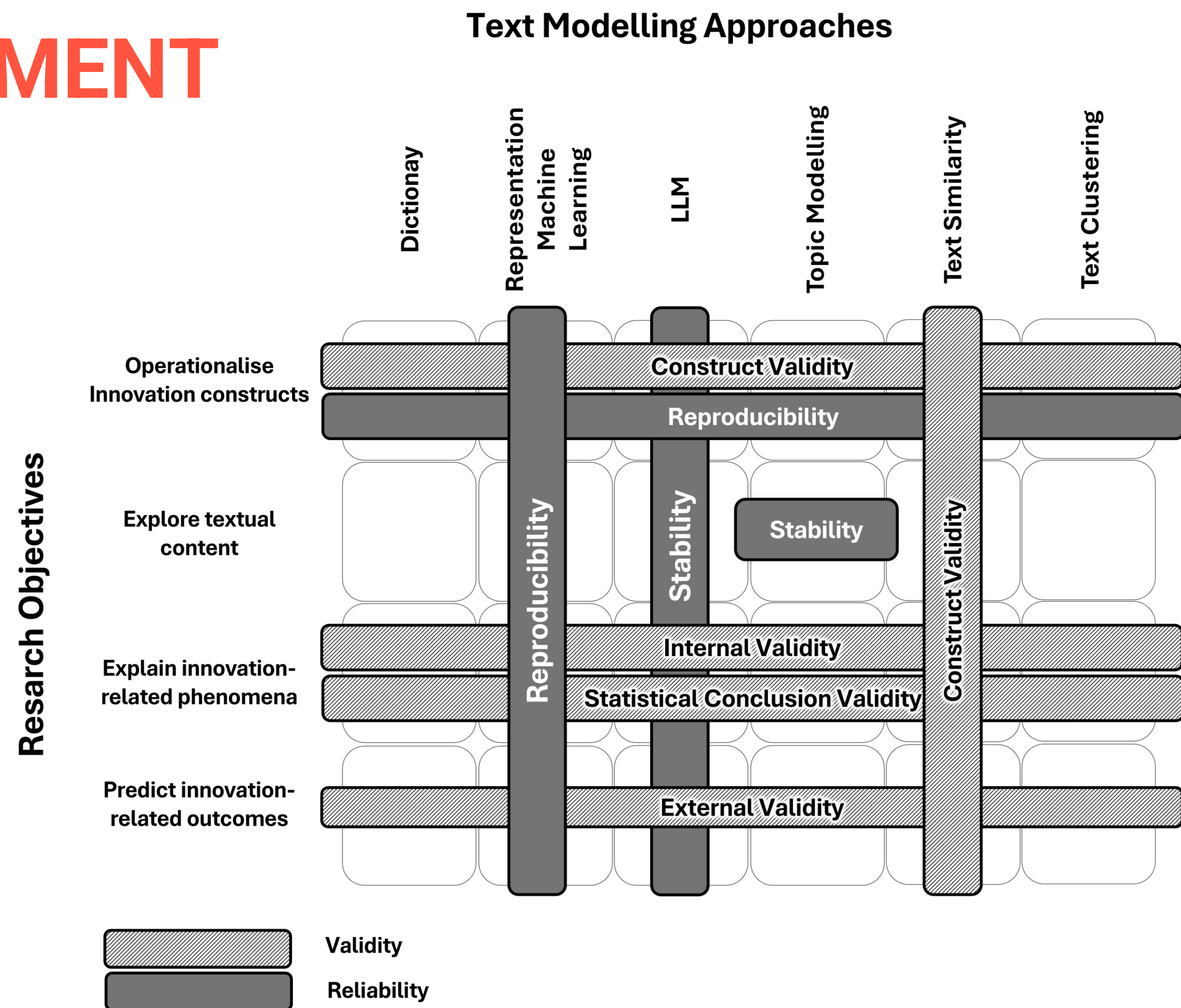
4



RIGOR IN MANAGEMENT

Guidelines

Which **validity** and **reliability** dimensions should be considered according to the specific **research context** of a text-analysis study.



RIGOR IN MANAGEMENT

Guidelines

Rigor Dimension	Strategy	No. of papers	% of papers
Construct Validity	Define the construct before operationalization	29	16.67
	Assess conceptual alignment between operationalization and construct	24	13.79
	Compare multiple operationalizations of the construct	57	32.76
	Ensure construct distinctiveness	5	2.87
	Minimize construct confounding	16	9.20
	Assess relationships expected with other constructs	26	14.94
Statistical Conclusion Validity	Increase Statistical Power	4	2.30
	Verify statistical test assumptions	5	2.87
	Account for unit heterogeneity	10	5.75
	Control error rates in multiple testing	2	1.15
Internal Validity	Establish temporal precedence	8	4.60
	Control for selection bias	25	14.37
	Control for history effects	9	5.17
	Assess mortality effects	2	1.15
	Assess maturation effects	3	1.72
External Validity	Generalise over different samples	13	7.47
	Generalise to observations over time	4	2.30
	Generalise to other aspects of a construct	7	4.02
	Generalise to real-world context	29	16.67
Stability	Control algorithmic randomness	3	1.72
	Control computational environment	1	0.57
Reproducibility	Ensure data accessibility	17	9.77
	Ensure code availability	17	9.77

Text Analysis in Innovation Research Canvas

Start by defining your **research context** then identify the **rigor strategies** you will use to address validity and reliability concerns.

1

Research Objectives

State one clear research objective for your study.

☐ Operationalise an innovation construct

☐ Exploratory content analysis

☐ Explanatory model

☐ Predict outcomes

2

Innovation Construct

Define the construct that your text analysis method aims to capture.

3

Predefined or Customised Modelling

Specify whether your text analysis method relies on a predefined or a customised approach.

☐ Predefined

☐ Customized

Reference supporting the predefined operationalisation:

4

Text Modelling Approach

Choose the text modelling approach you will use to operationalise the construct in your study.

☐ Dictionary-based

☐ Topic Modelling

☐ Machine Learning

☐ Text Similarity

☐ Large Language Model

☐ Text Clustering

5

Validity

Validity refers to the degree to which the conclusions drawn from a study can be considered accurate and credible. Based on your research context: (1) Check the validity dimensions you need to prioritise; (2) Describe the rigor strategies you plan to apply.

Statistical Conclusion Validity

Internal Validity

External Validity

Construct Validity

6

Reliability

Reliability refers to the consistency of those inferences under similar conditions. Based on your research context: (1) Check the validity dimensions you need to prioritise; (2) Describe the rigor strategies you plan to apply.

Stability

Reproducibility

Guides
researchers in
defining their
research context
and selecting the
rigor strategies
needed to
address the
relevant validity
and reliability
dimensions.



**Thanks for the
attention!**

Vito Giordano

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